

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$f(x) = 27(1.20)^{\frac{x}{3}}$$

For the given function f , the value of $f(x)$ increases by $p\%$ for every increase of x by 6. What is the value of p ?

Correct Answer: 44

Rationale

The correct answer is 44. For a function defined by an equation of the form $f(x) = a\left(1 + \frac{r}{100}\right)^{\frac{x}{k}}$, where a , r , and k are positive constants, the value of $f(x)$ increases by $r\%$ for every increase of x by k . The given equation can be rewritten as $f(x) = 27(1.20)^{\frac{2x}{6}}$, or $f(x) = 27\left((1.2)^2\right)^{\frac{x}{6}}$, which is equivalent to $f(x) = 27(1.44)^{\frac{x}{6}}$, or $f(x) = 27\left(1 + \frac{44}{100}\right)^{\frac{x}{6}}$. This equation is of the form $f(x) = a\left(1 + \frac{r}{100}\right)^{\frac{x}{k}}$, where $a = 27$, $r = 44$, and $k = 6$. It follows that the value of $f(x)$ increases by 44% for every increase of x by 6. Therefore, the value of p is 44.

Alternate approach: It's given that $f(x) = 27(1.20)^{\frac{x}{3}}$. An increase in the value of x by 6 can be represented by the expression $x + 6$. The value of $f(x + 6)$ can be found by substituting $x + 6$ for x in the equation $f(x) = 27(1.20)^{\frac{x}{3}}$, which yields $f(x + 6) = 27(1.20)^{\frac{x+6}{3}}$. By the product property of exponents, this equation is equivalent to $f(x + 6) = 27(1.20)^{\frac{x}{3}} \cdot (1.20)^{\frac{6}{3}}$, or $f(x + 6) = 27(1.20)^{\frac{x}{3}} \cdot (1.20)^2$, which can be rewritten as $f(x + 6) = 27(1.20)^{\frac{x}{3}} \cdot (1.44)$. Since $f(x) = 27(1.20)^{\frac{x}{3}}$, it follows that $f(x + 6) = f(x) \cdot (1.44)$. Therefore, for any value of x , the value of $f(x + 6)$ is 1.44 times, or 44% greater than, the value of $f(x)$. It's given that the value of $f(x)$ increases by $p\%$ for every increase in the value of x by 6. It follows that the value of p is 44.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$y = 9\left(\frac{a}{7}\right)^{x+c} - b$$

How many times does the graph of the given equation in the xy -plane cross the x -axis, where a , b , and c are positive constants such that $a > 7$ and $b > c$?

Answer

- A. Zero
- B. One
- C. Two
- D. Three

Correct Answer: B

Rationale

Choice B is correct. It’s given that $y = 9\left(\frac{a}{7}\right)^{x+c} - b$, where a , b , and c are positive constants such that $a > 7$ and $b > c$. The graph of any equation in the xy -plane crosses the x -axis when there is a point on the graph for which the value of y is equal to 0, and the value of y changes sign when the graph passes through that point. Since $a > 7$, the base $\frac{a}{7}$ is greater than 1, so the expression $9\left(\frac{a}{7}\right)^{x+c} - b$ increases as x increases. Since b is positive, there is a point on the graph of $y = 9\left(\frac{a}{7}\right)^{x+c} - b$ for which the value of y is equal to 0. When the graph passes through this point, the value of y changes sign from negative to positive. Therefore, the graph of the given equation crosses the x -axis exactly one time.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 128c75e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function g is defined by $g(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $g(15a)$ and $g(7a)$?

Correct Answer: 609

Rationale

The correct answer is **609**. It's given that the function g is defined by $g(x) = \frac{|x|}{a} - 14$, where $a < 0$. Substituting $15a$ for x in function g yields $g(15a) = \frac{|15a|}{a} - 14$. This function can be rewritten as $g(15a) = \frac{15|a|}{a} - 14$, or $g(15a) = 15\left(\frac{|a|}{a}\right) - 14$. Since $a < 0$, it follows that $\frac{|a|}{a} = -1$. Substituting -1 for $\frac{|a|}{a}$ in $g(15a) = 15\left(\frac{|a|}{a}\right) - 14$ yields $g(15a) = 15(-1) - 14$, or $g(15a) = -29$. Similarly, substituting $7a$ for x in function g yields $g(7a) = \frac{|7a|}{a} - 14$. This function can be rewritten as $g(7a) = \frac{7|a|}{a} - 14$, or $g(7a) = 7\left(\frac{|a|}{a}\right) - 14$. Since $a < 0$, it again follows that $\frac{|a|}{a} = -1$. Substituting -1 for $\frac{|a|}{a}$ in $g(7a) = 7\left(\frac{|a|}{a}\right) - 14$ yields $g(7a) = 7(-1) - 14$, or $g(7a) = -21$. Therefore, $g(15a) = -29$ and $g(7a) = -21$. Thus, the product of $g(15a)$ and $g(7a)$ is $(-29)(-21)$, or **609**.

Question ID: 91e7ea5e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$h(x) = 2(x - 4)^2 - 32$$

The quadratic function h is defined as shown. In the xy-plane, the graph of $y = h(x)$ intersects the x-axis at the points $(0,0)$ and $(t,0)$, where t is a constant. What is the value of t ?

Answer

- A. 1
- B. 2
- C. 4
- D. 8

Correct Answer: D

Rationale

Choice D is correct. It’s given that the graph of $y = h(x)$ intersects the x-axis at $(0,0)$ and $(t,0)$, where t is a constant. Since this graph intersects the x-axis when $y = 0$ or when $h(x) = 0$, it follows that $h(0) = 0$ and $h(t) = 0$. If $h(t) = 0$, then $0 = 2(t - 4)^2 - 32$. Adding 32 to both sides of this equation yields $32 = 2(t - 4)^2$. Dividing both sides of this equation by 2 yields $16 = (t - 4)^2$. Taking the square root of both sides of this equation yields $4 = t - 4$. Adding 4 to both sides of this equation yields $8 = t$. Therefore, the value of t is 8.

Choices A, B, and C are incorrect and may result from calculation errors.

Question ID: 3a9d60b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$2|4 - x| + 3|4 - x| = 25$

What is the positive solution to the given equation?

Correct Answer: 9

Rationale

The correct answer is 9. The given equation can be rewritten as $5|4 - x| = 25$. Dividing each side of this equation by 5 yields $|4 - x| = 5$. By the definition of absolute value, if $|4 - x| = 5$, then $4 - x = 5$ or $4 - x = -5$. Subtracting 4 from each side of the equation $4 - x = 5$ yields $-x = 1$. Dividing each side of this equation by -1 yields $x = -1$. Similarly, subtracting 4 from each side of the equation $4 - x = -5$ yields $-x = -9$. Dividing each side of this equation by -1 yields $x = 9$. Therefore, since the two solutions to the given equation are -1 and 9, the positive solution to the given equation is 9.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

An auditorium has seats for 1,800 people. Tickets to attend a show at the auditorium currently cost \$4.00. For each \$1.00 increase to the ticket price, 100 fewer tickets will be sold. This situation can be modeled by the equation $y = -100x^2 + 1,400x + 7,200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. If this equation is graphed in the xy -plane, at what value of x is the maximum of the graph?

Answer

- A. 4
- B. 7
- C. 14
- D. 18

Correct Answer: B

Rationale

Choice B is correct. It’s given that the situation can be modeled by the equation $y = -100x^2 + 1,400x + 7,200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. Since the coefficient of the x^2 term is negative, the graph of this equation in the xy -plane opens downward and reaches its maximum value at its vertex. If a quadratic equation in the form $y = ax^2 + bx + c$, where a , b , and c are constants, is graphed in the xy -plane, the x -coordinate of the vertex is equal to $-\frac{b}{2a}$. For the equation $y = -100x^2 + 1,400x + 7,200$, $a = -100$, $b = 1,400$, and $c = 7,200$. It follows that the x -coordinate of the vertex is $-\frac{1,400}{2(-100)}$, or 7. Therefore, if the given equation is graphed in the xy -plane, the maximum of the graph occurs at an x -value of 7.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

In the xy -plane, a line with equation $2y = 4.5$ intersects a parabola at exactly one point. If the parabola has equation $y = -4x^2 + bx$, where b is a positive constant, what is the value of b ?

Correct Answer: 6

Rationale

The correct answer is **6**. It's given that a line with equation $2y = 4.5$ intersects a parabola with equation $y = -4x^2 + bx$, where b is a positive constant, at exactly one point in the xy -plane. It follows that the system of equations consisting of $2y = 4.5$ and $y = -4x^2 + bx$ has exactly one solution. Dividing both sides of the equation of the line by **2** yields $y = 2.25$. Substituting **2.25** for y in the equation of the parabola yields $2.25 = -4x^2 + bx$. Adding $4x^2$ and subtracting bx from both sides of this equation yields $4x^2 - bx + 2.25 = 0$. A quadratic equation in the form of $ax^2 + bx + c = 0$, where a , b , and c are constants, has exactly one solution when the discriminant, $b^2 - 4ac$, is equal to zero. Substituting **4** for a and **2.25** for c in the expression $b^2 - 4ac$ and setting this expression equal to **0** yields $b^2 - 4(4)(2.25) = 0$, or $b^2 - 36 = 0$. Adding **36** to each side of this equation yields $b^2 = 36$. Taking the square root of each side of this equation yields $b = \pm 6$. It's given that b is positive, so the value of b is **6**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$f(x) = 9,000(0.66)^x$$

The given function f models the number of advertisements a company sent to its clients each year, where x represents the number of years since **1997**, and $0 \leq x \leq 5$. If $y = f(x)$ is graphed in the xy -plane, which of the following is the best interpretation of the y -intercept of the graph in this context?

Answer

- A. The minimum estimated number of advertisements the company sent to its clients during the **5** years was **1,708**.
- B. The minimum estimated number of advertisements the company sent to its clients during the **5** years was **9,000**.
- C. The estimated number of advertisements the company sent to its clients in **1997** was **1,708**.
- D. The estimated number of advertisements the company sent to its clients in **1997** was **9,000**.

Correct Answer: D

Rationale

Choice D is correct. The y -intercept of a graph in the xy -plane is the point where $x = 0$. For the given function f , the y -intercept of the graph of $y = f(x)$ in the xy -plane can be found by substituting **0** for x in the equation $y = 9,000(0.66)^x$, which gives $y = 9,000(0.66)^0$. This is equivalent to $y = 9,000(1)$, or $y = 9,000$. Therefore, the y -intercept of the graph of $y = f(x)$ is **(0, 9,000)**. It's given that the function f models the number of advertisements a company sent to its clients each year. Therefore, $f(x)$ represents the estimated number of advertisements the company sent to its clients each year. It's also given that x represents the number of years since **1997**. Therefore, $x = 0$ represents **0** years since **1997**, or **1997**. Thus, the best interpretation of the y -intercept of the graph of $y = f(x)$ is that the estimated number of advertisements the company sent to its clients in **1997** was **9,000**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$x - y = 1$
 $x + y = x^2 - 3$

Which ordered pair is a solution to the system of equations above?

Answer

- A. $(1 + \sqrt{3}, \sqrt{3})$
- B. $(\sqrt{3}, -\sqrt{3})$
- C. $(1 + \sqrt{5}, \sqrt{5})$
- D. $(\sqrt{5}, -1 + \sqrt{5})$

Correct Answer: A

Rationale

Choice A is correct. The solution to the given system of equations can be found by solving the first equation for x , which gives $x = y + 1$, and substituting that value of x into the second equation which gives $y + 1 + y = (y + 1)^2 - 3$. Rewriting this equation by adding like terms and expanding $(y + 1)^2$ gives $2y + 1 = y^2 + 2y - 2$. Subtracting $2y$ from both sides of this equation gives $1 = y^2 - 2$. Adding 2 to both sides of this equation gives $3 = y^2$. Therefore, it follows that $y = \pm\sqrt{3}$. Substituting $\sqrt{3}$ for y in the first equation yields $x - \sqrt{3} = 1$. Adding $\sqrt{3}$ to both sides of this equation yields $x = 1 + \sqrt{3}$. Therefore, the ordered pair $(1 + \sqrt{3}, \sqrt{3})$ is a solution to the given system of equations.

Choice B is incorrect. Substituting $\sqrt{3}$ for x and $-\sqrt{3}$ for y in the first equation yields $\sqrt{3} - (-\sqrt{3}) = 1$, or $2\sqrt{3} = 1$, which isn't a true statement. Choice C is incorrect. Substituting $1 + \sqrt{5}$ for x and $\sqrt{5}$ for y in the second equation yields $(1 + \sqrt{5}) + \sqrt{5} = (1 + \sqrt{5})^2 - 3$, or $1 + 2\sqrt{5} = 2\sqrt{5} + 3$, which isn't a true statement. Choice D is incorrect. Substituting $\sqrt{5}$ for x and $(-1 + \sqrt{5})$ for y in the second equation yields $\sqrt{5} + (-1 + \sqrt{5}) = (\sqrt{5})^2 - 3$, or $2\sqrt{5} - 1 = 2$, which isn't a true statement.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$(ax + 3)(5x^2 - bx + 4) = 20x^3 - 9x^2 - 2x + 12$

The equation above is true for all x, where a and b are constants. What is the value of ab ?

Answer

- A. 18
- B. 20
- C. 24
- D. 40

Correct Answer: C

Rationale

Choice C is correct. If the equation is true for all x, then the expressions on both sides of the equation will be equivalent. Multiplying the polynomials on the left-hand side of the equation gives $5ax^3 - abx^2 + 4ax + 15x^2 - 3bx + 12$. On the right-hand side of the equation, the only x^2 -term is $-9x^2$. Since the expressions on both sides of the equation are equivalent, it follows that $-abx^2 + 15x^2 = -9x^2$, which can be rewritten as $(-ab + 15)x^2 = -9x^2$. Therefore, $-ab + 15 = -9$, which gives $ab = 24$.

Choice A is incorrect. If $ab = 18$, then the coefficient of x^2 on the left-hand side of the equation would be $-18 + 15 = -3$, which doesn't equal the coefficient of x^2 , -9 , on the right-hand side. Choice B is incorrect. If $ab = 20$, then the coefficient of x^2 on the left-hand side of the equation would be $-20 + 15 = -5$, which doesn't equal the coefficient of x^2 , -9 , on the right-hand side. Choice D is incorrect. If $ab = 40$, then the coefficient of x^2 on the left-hand side of the equation would be $-40 + 15 = -25$, which doesn't equal the coefficient of x^2 , -9 , on the right-hand side.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

Which of the following expressions is(are) a factor of $3x^2 + 20x - 63$?

- I. $x - 9$
- II. $3x - 7$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Correct Answer: B

Rationale

Choice B is correct. The given expression can be factored by first finding two values whose sum is **20** and whose product is **3(−63)**, or **−189**. Those two values are **27** and **−7**. It follows that the given expression can be rewritten as $3x^2 + 27x - 7x - 63$. Since the first two terms of this expression have a common factor of **3x** and the last two terms of this expression have a common factor of **−7**, this expression can be rewritten as $3x(x + 9) - 7(x + 9)$. Since the two terms of this expression have a common factor of **(x + 9)**, it can be rewritten as $(3x - 7)(x + 9)$. Therefore, expression II, **3x − 7**, is a factor of $3x^2 + 20x - 63$, but expression I, **x − 9**, is not a factor of $3x^2 + 20x - 63$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

If $\frac{\sqrt{x^5}}{\sqrt[3]{x^4}} = x^{\frac{a}{b}}$ for all positive values of x, what is the value of $\frac{a}{b}$?

Rationale

The correct answer is $\frac{7}{6}$. The value of $\frac{a}{b}$ can be found by first rewriting the left-hand side of the given equation as $\frac{x^{\frac{5}{2}}}{x^{\frac{4}{3}}}$. Using the properties of exponents, this expression can be rewritten as $x^{\left(\frac{5}{2} - \frac{4}{3}\right)}$. This expression can be rewritten by subtracting the fractions in the exponent, which yields $x^{\frac{7}{6}}$. Thus, $\frac{a}{b}$ is $\frac{7}{6}$. Note that 7/6, 1.166, and 1.167 are examples of ways to enter a correct answer.

Question ID: b8f13a3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

Function f is defined by $f(x) = -a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x) - 12$ has a y -intercept at $(0, -\frac{75}{7})$. The product of a and b is $\frac{320}{7}$. What is the value of a ?

Correct Answer: 20

Rationale

The correct answer is **20**. It's given that $f(x) = -a^x + b$. Substituting $-a^x + b$ for $f(x)$ in the equation $y = f(x) - 12$ yields $y = -a^x + b - 12$. It's given that the y -intercept of the graph of $y = f(x) - 12$ is $(0, -\frac{75}{7})$. Substituting 0 for x and $-\frac{75}{7}$ for y in the equation $y = -a^x + b - 12$ yields $-\frac{75}{7} = -a^0 + b - 12$, which is equivalent to $-\frac{75}{7} = -1 + b - 12$, or $-\frac{75}{7} = b - 13$. Adding 13 to both sides of this equation yields $\frac{16}{7} = b$. It's given that the product of a and b is $\frac{320}{7}$, or $ab = \frac{320}{7}$. Substituting $\frac{16}{7}$ for b in this equation yields $(a)(\frac{16}{7}) = \frac{320}{7}$. Dividing both sides of this equation by $\frac{16}{7}$ yields $a = 20$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$r^2 + qr = 2r - 55$

In the given equation, q is an integer constant. The given equation has no real solutions. What is the largest possible value of q ?

Correct Answer: 16

Rationale

The correct answer is **16**. A quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has no real solution if and only if its discriminant, $b^2 - 4ac$, is negative. It's given that in the equation $r^2 + qr = 2r - 55$, q is an integer constant. Subtracting $2r$ and adding 55 to both sides of this equation yields $r^2 + qr - 2r + 55 = 0$, or $r^2 + (q - 2)r + 55 = 0$. In this equation, $a = 1$, $b = q - 2$, and $c = 55$. Substituting 1 for a , $q - 2$ for b , and 55 for c in the discriminant expression yields $(q - 2)^2 - 4(1)(55)$, or $(q - 2)^2 - 220$. Since the value of the discriminant must be negative, it follows that $(q - 2)^2 - 220 < 0$, or $(q - 2)^2 < 220$. This is equivalent to $|q - 2| < \sqrt{220}$, or $-\sqrt{220} < q - 2 < \sqrt{220}$. Adding 2 to each part of this inequality yields $2 - \sqrt{220} < q < 2 + \sqrt{220}$. Since $\sqrt{220}$ is approximately **14.83**, it follows that $-12.83 < q < 16.83$. Since q is an integer, the largest possible value of q is **16**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$\frac{1}{x^2+10x+25} = 4$$

If x is a solution to the given equation, which of the following is a possible value of $x+5$?

- Answer**
- A. $\frac{1}{2}$
- B. $\frac{5}{2}$
- C. $\frac{9}{2}$
- D. $\frac{11}{2}$

Correct Answer: A

Rationale

Choice A is correct. The given equation can be rewritten as $\frac{1}{(x+5)^2} = 4$. Multiplying both sides of this equation by $(x+5)^2$ yields $1 = 4(x+5)^2$. Dividing both sides of this equation by 4 yields $\frac{1}{4} = (x+5)^2$. Taking the square root of both sides of this equation yields $\frac{1}{2} = x+5$ or $-\frac{1}{2} = x+5$. Therefore, a possible value of $x+5$ is $\frac{1}{2}$.

Choices B, C, and D are incorrect and may result from computational or conceptual errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The expression $\frac{29x+102}{x(x+51)}$ is equivalent to $\frac{p}{x} + \frac{w}{x+51}$, where p and w are constants. What is the value of w ?

Correct Answer: 27

Rationale

The correct answer is **27**. The given expression $\frac{p}{x} + \frac{w}{x+51}$ can be rewritten with the common denominator $x(x+51)$. To add the two terms of this expression, $\frac{p}{x}$ can be rewritten as $\frac{p(x+51)}{x(x+51)}$ and $\frac{w}{(x+51)}$ can be rewritten as $\frac{wx}{x(x+51)}$. Therefore, the expression $\frac{p}{x} + \frac{w}{(x+51)}$ can be rewritten as $\frac{p(x+51)}{x(x+51)} + \frac{wx}{x(x+51)}$, which is equivalent to $\frac{px+51p+wx}{x(x+51)}$, or $\frac{(p+w)x+51p}{x(x+51)}$. Since $\frac{29x+102}{x(x+51)}$ is equivalent to $\frac{(p+w)x+51p}{x(x+51)}$ and these expressions have equivalent denominators, these expressions must also have equivalent numerators. It follows that $29x + 102 = (p + w)x + 51p$. In this equation, the x-coefficients must be equal and the constant terms must be equal. Therefore, $p + w = 29$ and $51p = 102$. Dividing both sides of $51p = 102$ by 51 yields $p = 2$. Substituting 2 for p in $p + w = 29$ yields $2 + w = 29$. Subtracting 2 from both sides of this equation yields $w = 27$. Therefore, the value of w is **27**.

Question ID: f2f3fa00

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

During a 5-second time interval, the average acceleration a , in meters per second squared, of an object with an initial velocity of 12 meters per second is defined by the equation $a = \frac{v_f - 12}{5}$, where v_f is the final velocity of the object in meters per second. If the equation is rewritten in the form $v_f = xa + y$, where x and y are constants, what is the value of x ?

Rationale

The correct answer is 5. The given equation can be rewritten in the form $v_f = xa + y$, like so:

$$a = \frac{v_f - 12}{5}$$

$$v_f - 12 = 5a$$

$$v_f = 5a + 12$$

It follows that the value of x is 5 and the value of y is 12.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question
 $f(x) = -500x^2 + 25,000x$

The revenue $f(x)$, in dollars, that a company receives from sales of a product is given by the function f above, where x is the unit price, in dollars, of the product. The graph of $y = f(x)$ in the xy -plane intersects the x -axis at 0 and a . What does a represent?

- Answer**
- A. The revenue, in dollars, when the unit price of the product is \$0
 - B. The unit price, in dollars, of the product that will result in maximum revenue
 - C. The unit price, in dollars, of the product that will result in a revenue of \$0
 - D. The maximum revenue, in dollars, that the company can make

Correct Answer: C

Rationale

Choice C is correct. By definition, the y -value when a function intersects the x -axis is 0. It's given that the graph of the function intersects the x -axis at 0 and a , that x is the unit price, in dollars, of a product, and that $f(x)$, where $y = f(x)$, is the revenue, in dollars, that a company receives from the sales of the product. Since the value of a occurs when $y = 0$, a is the unit price, in dollars, of the product that will result in a revenue of \$0.

Choice A is incorrect. The revenue, in dollars, when the unit price of the product is \$0 is represented by $f(x)$, when $x = 0$. Choice B is incorrect. The unit price, in dollars, of the product that will result in maximum revenue is represented by the x -coordinate of the maximum of f . Choice D is incorrect. The maximum revenue, in dollars, that the company can make is represented by the y -coordinate of the maximum of f .

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$\frac{2}{x-2} + \frac{3}{x+5} = \frac{rx+t}{(x-2)(x+5)}$$

The equation above is true for all $x > 2$, where r and t are positive constants. What is the value of rt ?

Answer

- A. -20
- B. 15
- C. 20
- D. 60

Correct Answer: C

Rationale

Choice C is correct. To express the sum of the two rational expressions on the left-hand side of the equation as the single rational expression on the right-hand side of the equation, the expressions on the left-hand side must have the same denominator. Multiplying the first expression by $\frac{x+5}{x-5}$ results in $\frac{2(x+5)}{(x-2)(x+5)}$, and multiplying the second expression by $\frac{x-2}{x-2}$ results in $\frac{3(x-2)}{(x-2)(x+5)}$, so the given equation can be rewritten as $\frac{2(x+5)}{(x-2)(x+5)} + \frac{3(x-2)}{(x-2)(x+5)} = \frac{rx+t}{(x-2)(x+5)}$, or $\frac{2x+10}{(x-2)(x+5)} + \frac{3x-6}{(x-2)(x+5)} = \frac{rx+t}{(x-2)(x+5)}$. Since the two rational expressions on the left-hand side of the equation have the same denominator as the rational expression on the right-hand side of the equation, it follows that $(2x+10)+(3x-6)=rx+t$. Combining like terms on the left-hand side yields $5x+4=rx+t$, so it follows that $r=5$ and $t=4$. Therefore, the value of rt is $(5)(4)=20$.

Choice A is incorrect and may result from an error when determining the sign of either r or t . Choice B is incorrect and may result from not distributing the 2 and 3 to their respective terms in $\frac{2(x+5)}{(x-2)(x+5)} + \frac{3(x-2)}{(x-2)(x+5)} = \frac{rx+t}{(x-2)(x+5)}$. Choice D is incorrect and may result from a calculation error.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

Growth of a Culture of Bacteria

Day	Number of bacteria per milliliter at end of day
1	2.5×10^5
2	5.0×10^5
3	1.0×10^6

A culture of bacteria is growing at an exponential rate, as shown in the table above. At this rate, on which day would the number of bacteria per milliliter reach 5.12×10^8 ?

- Answer
- A. Day 5
 - B. Day 9
 - C. Day 11
 - D. Day 12

Correct Answer: D

Rationale

Choice D is correct. The number of bacteria per milliliter is doubling each day. For example, from day 1 to day 2, the number of bacteria increased from 2.5×10^5 to 5.0×10^5 . At the end of day 3 there are 10^6 bacteria per milliliter. At the end of day 4, there will be $10^6 \times 2$ bacteria per milliliter. At the end of day 5, there will be $(10^6 \times 2) \times 2$, or $10^6 \times (2^2)$ bacteria per milliliter, and so on. At the end of day d, the number of bacteria will be $10^6 \times (2^{d-3})$. If the number of bacteria per milliliter will reach 5.12×10^8 at the end of day d, then the equation $10^6 \times (2^{d-3}) = 5.12 \times 10^8$ must hold. Since 5.12×10^8 can be rewritten as 512×10^6 , the equation is equivalent to $2^{d-3} = 512$. Rewriting 512 as 2^9 gives $d - 3 = 9$, so $d = 12$. The number of bacteria per milliliter would reach 5.12×10^8 at the end of day 12.

Choices A, B, and C are incorrect. Given the growth rate of the bacteria, the number of bacteria will not reach 5.12×10^8 per milliliter by the end of any of these days.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$\sqrt[5]{70n}\left(\sqrt[6]{70n}\right)^2$$

For what value of x is the given expression equivalent to $(70n)^{30x}$, where $n > 1$?

Correct Answer: .0177, .0178, 4/225

Rationale

The correct answer is $\frac{4}{225}$. An expression of the form $\sqrt[k]{a}$, where k is an integer greater than 1 and $a \geq 0$, is equivalent to $a^{\frac{1}{k}}$. Therefore, the given expression, where $n > 1$, is equivalent to $(70n)^{\frac{1}{5}}\left((70n)^{\frac{1}{6}}\right)^2$. Applying properties of exponents, this expression can be rewritten as $(70n)^{\frac{1}{5}}(70n)^{\frac{1}{6}\cdot 2}$, or $(70n)^{\frac{1}{5}}(70n)^{\frac{1}{3}}$, which can be rewritten as $(70n)^{\frac{1}{5}+\frac{1}{3}}$, or $(70n)^{\frac{8}{15}}$. It's given that the expression $\sqrt[5]{70n}\left(\sqrt[6]{70n}\right)^2$ is equivalent to $(70n)^{30x}$, where $n > 1$. It follows that $(70n)^{\frac{8}{15}}$ is equivalent to $(70n)^{30x}$. Therefore, $\frac{8}{15} = 30x$. Dividing both sides of this equation by 30 yields $\frac{8}{450} = x$, or $\frac{4}{225} = x$. Thus, the value of x for which the given expression is equivalent to $(70n)^{30x}$, where $n > 1$, is $\frac{4}{225}$. Note that 4/225, .0177, .0178, 0.017, and 0.018 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question
 $2x^2 - 2 = 2x + 3$

Which of the following is a solution to the equation above?

- Answer
- A. 2
- B. $1 - \sqrt{11}$
- C. $\frac{1}{2} + \sqrt{11}$
- D. $\frac{1 + \sqrt{11}}{2}$

Correct Answer: D

Rationale

Choice D is correct. A quadratic equation in the form $ax^2 + bx + c = 0$, where a, b, and c are constants, can be solved using the quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Subtracting $2x + 3$ from both sides of the given equation yields $2x^2 - 2x - 5 = 0$. Applying the quadratic formula, where $a = 2$, $b = -2$, and $c = -5$, yields $x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(2)(-5)}}{2(2)}$. This can be rewritten as $x = \frac{2 \pm \sqrt{44}}{4}$. Since $\sqrt{44} = \sqrt{2^2(11)}$, or $2\sqrt{11}$, the equation can be rewritten as $x = \frac{2 \pm 2\sqrt{11}}{4}$. Dividing 2 from both the numerator and denominator yields $\frac{1 + \sqrt{11}}{2}$ or $\frac{1 - \sqrt{11}}{2}$. Of these two solutions, only $\frac{1 + \sqrt{11}}{2}$ is present among the choices. Thus, the correct choice is D.

Choice A is incorrect and may result from a computational or conceptual error. Choice B is incorrect and may result from using $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{a}$ instead of $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ as the quadratic formula. Choice C is incorrect and may result from rewriting $\sqrt{44}$ as $4\sqrt{11}$ instead of $2\sqrt{11}$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = ax^2 + 4x + c$

In the given quadratic function, a and c are constants. The graph of $y = f(x)$ in the xy -plane is a parabola that opens upward and has a vertex at the point (h, k) , where h and k are constants. If $k < 0$ and $f(-9) = f(3)$, which of the following must be true?

- I. $c < 0$
- II. $a \geq 1$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Correct Answer: D

Rationale

Choice D is correct. It's given that the graph of $y = f(x)$ in the xy -plane is a parabola with vertex (h, k) . If $f(-9) = f(3)$, then for the graph of $y = f(x)$, the point with an x -coordinate of -9 and the point with an x -coordinate of 3 have the same y -coordinate. In the xy -plane, a parabola is a symmetric graph such that when two points have the same y -coordinate, these points are equidistant from the vertex, and the x -coordinate of the vertex is halfway between the x -coordinates of these two points. Therefore, for the graph of $y = f(x)$, the points with x -coordinates -9 and 3 are equidistant from the vertex, (h, k) , and h is halfway between -9 and 3 . The value that is halfway between -9 and 3 is $\frac{-9+3}{2}$, or -3 . Therefore, $h = -3$. The equation defining f can also be written in vertex form, $f(x) = a(x - h)^2 + k$. Substituting -3 for h in this equation yields $f(x) = a(x - (-3))^2 + k$, or $f(x) = a(x + 3)^2 + k$. This equation is equivalent to $f(x) = a(x^2 + 6x + 9) + k$, or $f(x) = ax^2 + 6ax + 9a + k$. Since $f(x) = ax^2 + 4x + c$, it follows that $6a = 4$ and $9a + k = c$. Dividing both sides of the equation $6a = 4$ by 6 yields $a = \frac{4}{6}$, or $a = \frac{2}{3}$. Since $\frac{2}{3} < 1$, it's not true that $a \geq 1$. Therefore, statement II isn't true. Substituting $\frac{2}{3}$ for a in the equation $9a + k = c$ yields $9(\frac{2}{3}) + k = c$, or $6 + k = c$. Subtracting 6 from both sides of this equation yields $k = c - 6$. If $k < 0$, then $c - 6 < 0$, or $c < 6$. Since c could be any value less than 6 , it's not necessarily true that $c < 0$. Therefore, statement I isn't necessarily true. Thus, neither I nor II must be true.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: f5aa5040

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

In the xy -plane, a line with equation $2y = c$ for some constant c intersects a parabola at exactly one point. If the parabola has equation $y = -2x^2 + 9x$, what is the value of c ?

Correct Answer: 20.25, 81/4

Rationale

The correct answer is $\frac{81}{4}$. The given linear equation is $2y = c$. Dividing both sides of this equation by 2 yields $y = \frac{c}{2}$. Substituting $\frac{c}{2}$ for y in the equation of the parabola yields $\frac{c}{2} = -2x^2 + 9x$. Adding $2x^2$ and $-9x$ to both sides of this equation yields $2x^2 - 9x + \frac{c}{2} = 0$. Since it's given that the line and the parabola intersect at exactly one point, the equation $2x^2 - 9x + \frac{c}{2} = 0$ must have exactly one solution. An equation of the form $Ax^2 + Bx + C = 0$, where A , B , and C are constants, has exactly one solution when the discriminant, $B^2 - 4AC$, is equal to 0 . In the equation $2x^2 - 9x + \frac{c}{2} = 0$, where $A = 2$, $B = -9$, and $C = \frac{c}{2}$, the discriminant is $(-9)^2 - 4(2)(\frac{c}{2})$. Setting the discriminant equal to 0 yields $(-9)^2 - 4(2)(\frac{c}{2}) = 0$, or $81 - 4c = 0$. Adding $4c$ to both sides of this equation yields $81 = 4c$. Dividing both sides of this equation by 4 yields $c = \frac{81}{4}$. Note that 81/4 and 20.25 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The expression $(3x - 23)(19x + 6)$ is equivalent to the expression $ax^2 + bx + c$, where a , b , and c are constants. What is the value of b ?

Correct Answer: -419

Rationale

The correct answer is -419 . It's given that the expression $(3x - 23)(19x + 6)$ is equivalent to the expression $ax^2 + bx + c$, where a , b , and c are constants. Applying the distributive property to the given expression, $(3x - 23)(19x + 6)$, yields $(3x)(19x) + (3x)(6) - (23)(19x) - (23)(6)$, which can be rewritten as $57x^2 + 18x - 437x - 138$. Combining like terms yields $57x^2 - 419x - 138$. Since this expression is equivalent to $ax^2 + bx + c$, it follows that the value of b is -419 .

Question ID: 722de804

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$(x - 47)^2 = 1$

What is the sum of the solutions to the given equation?

Correct Answer: 94

Rationale

The correct answer is **94**. Taking the square root of each side of the given equation yields $x - 47 = 1$ or $x - 47 = -1$. Adding **47** to both sides of the equation $x - 47 = 1$ yields $x = 48$. Adding **47** to both sides of the equation $x - 47 = -1$ yields $x = 46$. Therefore, the sum of the solutions to the given equation is **48 + 46**, or **94**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$x^2 + y^2 = 36$$

$$y = mx + \frac{b}{4}$$

In the given system of equations, m and b are negative constants. In the xy -plane, the graphs of the equations in the given system intersect at the point $(-5, y)$, where $y < 0$. Which expression represents the value of b ?

Answer

- A. $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$
- B. $\frac{5m}{4} - \frac{\sqrt{11}}{4}$
- C. $-20m + 4\sqrt{11}$
- D. $20m - 4\sqrt{11}$

Correct Answer: D

Rationale

Choice D is correct. It's given that the graphs of $x^2 + y^2 = 36$ and $y = mx + \frac{b}{4}$, where m and b are negative constants, intersect at the point $(-5, y)$ in the xy -plane. It follows that this intersection point represents a solution to the given system of equations. Substituting -5 for x in the equation $x^2 + y^2 = 36$ yields $(-5)^2 + y^2 = 36$, or $25 + y^2 = 36$. Subtracting 25 from each side of this equation yields $y^2 = 11$. It's also given that $y < 0$, so taking the negative square root of each side of this equation yields $y = -\sqrt{11}$. Substituting -5 for x and $-\sqrt{11}$ for y in the equation $y = mx + \frac{b}{4}$ yields $-\sqrt{11} = m(-5) + \frac{b}{4}$, or $-\sqrt{11} = -5m + \frac{b}{4}$. Adding $5m$ to each side of this equation yields $5m - \sqrt{11} = \frac{b}{4}$. Multiplying each side of this equation by 4 yields $4(5m - \sqrt{11}) = b$. Applying the distributive property to the left-hand side of this equation yields $20m - 4\sqrt{11} = b$. Therefore, the expression $20m - 4\sqrt{11}$ represents the value of b .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

Which expression is equivalent to $\frac{4}{4x-5} - \frac{1}{x+1}$?

Answer

- A. $\frac{1}{(x+1)(4x-5)}$
- B. $\frac{3}{3x-6}$
- C. $-\frac{1}{(x+1)(4x-5)}$
- D. $\frac{9}{(x+1)(4x-5)}$

Correct Answer: D

Rationale

Choice D is correct. The expression $\frac{4}{4x-5} - \frac{1}{x+1}$ can be rewritten as $\frac{4}{4x-5} + \frac{(-1)}{x+1}$. To add the two terms of this expression, the terms can be rewritten with a common denominator. Since $\frac{x+1}{x+1} = 1$, the expression $\frac{4}{4x-5}$ can be rewritten as $\frac{(x+1)(4)}{(x+1)(4x-5)}$. Since $\frac{4x-5}{4x-5} = 1$, the expression $\frac{-1}{x+1}$ can be rewritten as $\frac{(4x-5)(-1)}{(4x-5)(x+1)}$. Therefore, the expression $\frac{4}{4x-5} + \frac{(-1)}{x+1}$ can be rewritten as $\frac{(x+1)(4)}{(x+1)(4x-5)} + \frac{(4x-5)(-1)}{(4x-5)(x+1)}$, which is equivalent to $\frac{(x+1)(4)+(4x-5)(-1)}{(x+1)(4x-5)}$. Applying the distributive property to each term of the numerator yields $\frac{(4x+4)+(-4x+5)}{(x+1)(4x-5)}$, or $\frac{(4x+(-4x))+(-4+5)}{(x+1)(4x-5)}$. Adding like terms in the numerator yields $\frac{9}{(x+1)(4x-5)}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: d8789a4c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$\frac{x^2 - c}{x - b}$$

In the expression above, b and c are positive integers. If the expression is equivalent to $x + b$ and $x \neq b$, which of the following could be the value of c ?

Answer

- A. 4
- B. 6
- C. 8
- D. 10

Correct Answer: A

Rationale

Choice A is correct. If the given expression is equivalent to $x + b$, then $\frac{x^2 - c}{x - b} = x + b$, where x isn't equal to b. Multiplying both sides of this equation by $x - b$ yields $x^2 - c = (x + b)(x - b)$. Since the right-hand side of this equation is in factored form for the difference of squares, the value of c must be a perfect square. Only choice A gives a perfect square for the value of c.

Choices B, C, and D are incorrect. None of these values of c produces a difference of squares. For example, when 6 is substituted for c in the given expression, the result is $\frac{x^2 - 6}{x - b}$. The expression $x^2 - 6$ can't be factored with integer values, and therefore $\frac{x^2 - 6}{x - b}$ isn't equivalent to $x + b$.

Question ID: 58b4c6f3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy -plane passes through the points $(0, 10)$ and $(2, 13)$. What is the value of ab ?

Answer

- A. 13
- B. 18
- C. 20
- D. 26

Correct Answer: B

Rationale

Choice B is correct. It's given that function h is defined by $h(x) = a^x + b$, where a and b are positive constants. It's also given that the graph of $y = h(x)$ in the xy -plane passes through the point $(0, 10)$. Substituting 0 for x and 10 for $h(x)$ in the equation $h(x) = a^x + b$ yields $10 = a^0 + b$. Since $a^0 = 1$, this equation can be rewritten as $10 = 1 + b$. Subtracting 1 from both sides of this equation yields $9 = b$. Substituting 9 for b in the equation $h(x) = a^x + b$ yields $h(x) = a^x + 9$. It's also given that the graph of $y = h(x)$ in the xy -plane passes through the point $(2, 13)$. Substituting 2 for x and 13 for $h(x)$ in the equation $h(x) = a^x + 9$ yields $13 = a^2 + 9$. Subtracting 9 from both sides of this equation yields $4 = a^2$. Taking the square root of both sides of this equation yields $2 = a$ and $-2 = a$. Since a is positive, it follows that $2 = a$. Since $9 = b$ and $2 = a$, it follows that the value of ab is $(2)(9)$, or 18 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A lab analyst observes a sample of a substance. An exponential model estimates that the mass, in grams, of the sample decreases by **24%** every **22.96** minutes. Which of the following equations could represent this model, where ***M*** is the estimated mass, in grams, of the sample ***t*** minutes after the lab analyst began observing the sample?

Answer

- A. $M = 100(0.24)^{t+22.96}$
- B. $M = 100(0.24)^{\frac{t}{22.96}}$
- C. $M = 100(0.76)^{t+22.96}$
- D. $M = 100(0.76)^{\frac{t}{22.96}}$

Correct Answer: D

Rationale

Choice D is correct. It’s given that the mass of a sample of a substance decreases by **24%** every **22.96** minutes. Therefore, after each **22.96**-minute interval the mass is multiplied by **1 — 0.24**, or **0.76**. It’s also given that ***t*** represents the number of minutes after the lab analyst began observing the sample; therefore, $\frac{t}{22.96}$ represents the number of **22.96**-minute intervals. The equation that could represent this model is ***M*** = **100(0.76) ^{$\frac{t}{22.96}$}** , where ***M*** is the estimated mass, in grams, of the sample ***t*** minutes after the lab analyst began observing it.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A park ranger hung squirrel houses each in the shape of a right rectangular prism for fox squirrels. Each house has a height of **11** inches. The length of each house's base is x inches, which is **1** inch more than the width of the house's base. Which function V gives the volume of each house, in cubic inches, in terms of the length of the house's base?

Answer

- A. $V(x) = 11x(x - 1)$
- B. $V(x) = 11x(x + 1)$
- C. $V(x) = x(x + 11)(x - 1)$
- D. $V(x) = x(x + 11)(x + 1)$

Correct Answer: A

Rationale

Choice A is correct. The volume of a prism is equal to the area of its base times its height. It's given that the length of each house's base is x inches and that this length is **1** inch more than the width, in inches, of the house's base. It follows that the width, in inches, of the house's base is $x - 1$. The area of a rectangle is the product of its length and its width. Therefore, the area of the base of the house is $x(x - 1)$ square inches. It's given that the height of each house is **11** inches. Therefore, the function V that gives the volume of each house, in cubic inches, in terms of the length of the house's base x is $V(x) = x(x - 1)11$, or $V(x) = 11x(x - 1)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = (x - 14)(x + 19)$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

Answer

- A. -266
- B. -19
- C. $-\frac{33}{2}$
- D. $-\frac{5}{2}$

Correct Answer: D

Rationale

Choice D is correct. It's given that $f(x) = (x - 14)(x + 19)$, which can be rewritten as $f(x) = x^2 + 5x - 266$. Since the coefficient of the x^2 -term is positive, the graph of $y = f(x)$ in the xy -plane opens upward and reaches its minimum value at its vertex. The x -coordinate of the vertex is the value of x such that $f(x)$ reaches its minimum. For an equation in the form $f(x) = ax^2 + bx + c$, where a , b , and c are constants, the x -coordinate of the vertex is $-\frac{b}{2a}$. For the equation $f(x) = x^2 + 5x - 266$, $a = 1$, $b = 5$, and $c = -266$. It follows that the x -coordinate of the vertex is $-\frac{5}{2(1)}$, or $-\frac{5}{2}$. Therefore, $f(x)$ reaches its minimum when the value of x is $-\frac{5}{2}$.

Alternate approach: The value of x for the vertex of a parabola is the x -value of the midpoint between the two x -intercepts of the parabola. Since it's given that $f(x) = (x - 14)(x + 19)$, it follows that the two x -intercepts of the graph of $y = f(x)$ in the xy -plane occur when $x = 14$ and $x = -19$, or at the points $(14, 0)$ and $(-19, 0)$. The midpoint between two points, (x_1, y_1) and (x_2, y_2) , is $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$. Therefore, the midpoint between $(14, 0)$ and $(-19, 0)$ is $(\frac{14+(-19)}{2}, \frac{0+0}{2})$, or $(-\frac{5}{2}, 0)$. It follows that $f(x)$ reaches its minimum when the value of x is $-\frac{5}{2}$.

Choice A is incorrect. This is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane.

Choice B is incorrect. This is one of the x -coordinates of the x -intercepts of the graph of $y = f(x)$ in the xy -plane.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$4x^2 - 5x - 5 = 0$

What is the greatest solution to the given equation?

Answer

- A. $\frac{5}{8} - \frac{\sqrt{105}}{8}$
- B. $\frac{5}{8} + \frac{\sqrt{105}}{8}$
- C. $\frac{5}{4} - \frac{\sqrt{105}}{4}$
- D. $\frac{5}{4} + \frac{\sqrt{105}}{4}$

Correct Answer: B

Rationale

Choice B is correct. The solutions to a quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be calculated using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The given equation is of this form, where $a = 4$, $b = -5$, and $c = -5$. Substituting these values into the quadratic formula yields $x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(4)(-5)}}{2(4)}$, or $x = \frac{5 \pm \sqrt{25 + 80}}{8}$. These solutions can be rewritten as $x = \frac{5 \pm \sqrt{105}}{8}$, or $x = \frac{5}{8} \pm \frac{\sqrt{105}}{8}$. Since $\frac{5}{8} + \frac{\sqrt{105}}{8}$ is greater than $\frac{5}{8} - \frac{\sqrt{105}}{8}$, it follows that the greatest solution to the given equation is $\frac{5}{8} + \frac{\sqrt{105}}{8}$.

Choice A is incorrect. This is the least, not the greatest, solution to the given equation.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is a quadratic function. In the xy -plane, the graph of $y = f(x)$ has a vertex at $(1, 9)$ and passes through the points $(2, 33)$ and $(-1, 105)$. What is the value of $f(-2) - f(0)$?

Answer

- A. 81
- B. 129
- C. 192
- D. 225

Correct Answer: C

Rationale

Choice C is correct. It's given that function f is a quadratic function, and in the xy -plane, the graph of $y = f(x)$ has a vertex at $(1, 9)$. In the xy -plane, the graph of a quadratic function with a vertex at (h, k) has an equation of the form $y = a(x - h)^2 + k$, where a is a constant. Therefore, function f can be written as $f(x) = a(x - 1)^2 + 9$. It's also given that the graph of $y = f(x)$ passes through the point $(2, 33)$. Substituting 2 for x and 33 for $f(x)$ in the equation $f(x) = a(x - 1)^2 + 9$ yields $33 = a(2 - 1)^2 + 9$, or $33 = a + 9$. Subtracting 9 from both sides of this equation yields $24 = a$. Substituting 24 for a in the equation $f(x) = a(x - 1)^2 + 9$ yields $f(x) = 24(x - 1)^2 + 9$. The value of $f(-2)$ can be found by substituting -2 for x in function f . Therefore, $f(-2) = 24(-2 - 1)^2 + 9$, which yields $f(-2) = 24(9) + 9$, or $f(-2) = 225$. The value of $f(0)$ can be found by substituting 0 for x in function f . Therefore, $f(0) = 24(0 - 1)^2 + 9$, which yields $f(0) = 24 + 9$, or $f(0) = 33$. It follows that $f(-2) - f(0) = 225 - 33$, or $f(-2) - f(0) = 192$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $f(-2)$, not the value of $f(-2) - f(0)$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The expression $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + c$?

Correct Answer: 15.5, 31/2

Rationale

The correct answer is 15.5. It's given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers. The expression $(3x^2 + a)(2x^2 + b)$ can be rewritten as $6x^4 + 3bx^2 + 2ax^2 + ab$, or $6x^4 + (3b + 2a)x^2 + ab$. Therefore, the expression $6x^4 + (3b + 2a)x^2 + ab$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3b + 2a = 31$ and $ab = 35$. Since a and b are positive integers and $ab = 35$, the possible values of a and b are 1 and 35, 5 and 7, 7 and 5, and 35 and 1, respectively. Of these options, the only pair that satisfies the equation $3b + 2a = 31$ is $a = 5$ and $b = 7$, since $3(7) + 2(5) = 31$. It's also given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. The expression $(3x^2 + c)(2x^2 + d)$ can be rewritten as $6x^4 + 3dx^2 + 2cx^2 + cd$, or $6x^4 + (3d + 2c)x^2 + cd$. Therefore, the expression $6x^4 + (3d + 2c)x^2 + cd$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3d + 2c = 31$ and $cd = 35$. Since it's given that c and d are positive nonintegers, it follows that c and d are not zero. Thus, dividing both sides of the equation $cd = 35$ by c yields $d = \frac{35}{c}$. Substituting $\frac{35}{c}$ for d in the equation $3d + 2c = 31$ yields $3\left(\frac{35}{c}\right) + 2c = 31$, or $\frac{105}{c} + 2c = 31$. Multiplying both sides of this equation by c yields $c\left(\frac{105}{c} + 2c\right) = c(31)$, or $105 + 2c^2 = 31c$. Subtracting $31c$ from both sides of this equation yields $105 + 2c^2 - 31c = 0$, or $2c^2 - 31c + 105 = 0$. This equation can be rewritten as $2c^2 - 10c - 21c + 105 = 0$. Since the first two terms of the expression $2c^2 - 10c - 21c + 105$ have a common factor of $2c$ and the last two terms of the expression $2c^2 - 10c - 21c + 105$ have a common factor of 21 , the expression $2c^2 - 10c - 21c + 105$ can be rewritten as $2c(c - 5) - 21(c - 5)$. Since each term of this expression has a common factor of $(c - 5)$, the expression $2c(c - 5) - 21(c - 5)$ can be rewritten as $(c - 5)(2c - 21)$. Thus, the equation $2c^2 - 31c + 105 = 0$ can be rewritten as $(c - 5)(2c - 21) = 0$. It follows that $c = 5$ or $c = \frac{21}{2}$. Since c is a positive noninteger, it follows that $c = \frac{21}{2}$, or $c = 10.5$. Substituting 5 for a and 10.5 for c into the expression $a + c$ yields $5 + 10.5$, or 15.5. Therefore, the value of $a + c$ is 15.5. Note that 15.5 and 31/2 are examples of ways to enter a correct answer.

Alternate approach: The correct answer is 15.5. It's given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers. The expression $(3x^2 + a)(2x^2 + b)$ can be rewritten as $6x^4 + 3bx^2 + 2ax^2 + ab$, or $6x^4 + (3b + 2a)x^2 + ab$. Therefore, the expression $6x^4 + (3b + 2a)x^2 + ab$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3b + 2a = 31$ and $ab = 35$. Since it's given that a and b are positive integers, it follows that a and b are not zero. Thus, dividing both sides of the equation $ab = 35$ by a yields $b = \frac{35}{a}$. Substituting $\frac{35}{a}$ for b in the equation $3b + 2a = 31$ yields $3\left(\frac{35}{a}\right) + 2a = 31$, or $\frac{105}{a} + 2a = 31$. Multiplying both sides of this equation by a yields $105 + 2a^2 = 31a$. Subtracting $31a$ from both sides of this equation yields $2a^2 - 31a + 105 = 0$. This equation can be rewritten as $2a^2 - 10a - 21a + 105 = 0$. Since the first two terms of the expression $2a^2 - 10a - 21a + 105$ have a common factor of $2a$ and the last two terms of the expression $2a^2 - 10a - 21a + 105$ have a common factor of 21 , the expression $2a^2 - 10a - 21a + 105$ can be rewritten as $2a(a - 5) - 21(a - 5)$. Since each term of this expression has a common factor of $(a - 5)$, the expression $2a(a - 5) - 21(a - 5)$ can be rewritten as $(a - 5)(2a - 21)$. Thus, the equation $2a^2 - 31a + 105 = 0$ can be rewritten as $(a - 5)(2a - 21) = 0$. It follows that $a = 5$ or $a = \frac{21}{2}$. Since a is a positive integer, it follows that $a = 5$. It's also given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. The expression $(3x^2 + c)(2x^2 + d)$ can be rewritten as $6x^4 + 3dx^2 + 2cx^2 + cd$, or $6x^4 + (3d + 2c)x^2 + cd$. Therefore, the expression $6x^4 + (3d + 2c)x^2 + cd$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3d + 2c = 31$ and $cd = 35$. Since it's given that c and d are positive nonintegers, it follows that c and d are not zero. Thus, dividing both sides of the equation $cd = 35$ by c yields $d = \frac{35}{c}$. Substituting $\frac{35}{c}$ for d in the equation $3d + 2c = 31$ yields $3\left(\frac{35}{c}\right) + 2c = 31$, or $\frac{105}{c} + 2c = 31$. Multiplying both sides of this equation by c yields $105 + 2c^2 = 31c$. Subtracting $31c$ from both sides of this equation yields $2c^2 - 31c + 105 = 0$. Since the equations $2a^2 - 31a + 105 = 0$ and $2c^2 - 31c + 105 = 0$ represent the same equation, it follows that a is equal to c . Therefore, $c = 5$ or $c = \frac{21}{2}$. Since c is a positive noninteger, it follows that $c = \frac{21}{2}$, or $c = 10.5$. Substituting 5 for a and 10.5 for c in the expression $a + c$ yields $5 + 10.5$, or 15.5. Therefore, the value of $a + c$ is 15.5. Note that 15.5 and 31/2 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$2x^2 - 4x = t$

In the equation above, t is a constant. If the equation has no real solutions, which of the following could be the value of t ?

Answer

- A. -3
- B. -1
- C. 1
- D. 3

Correct Answer: A

Rationale

Choice A is correct. The number of solutions to any quadratic equation in the form $ax^2 + bx + c = 0$, where a, b, and c are constants, can be found by evaluating the expression $b^2 - 4ac$, which is called the discriminant. If the value of $b^2 - 4ac$ is a positive number, then there will be exactly two real solutions to the equation. If the value of $b^2 - 4ac$ is zero, then there will be exactly one real solution to the equation. Finally, if the value of $b^2 - 4ac$ is negative, then there will be no real solutions to the equation.

The given equation $2x^2 - 4x = t$ is a quadratic equation in one variable, where t is a constant. Subtracting t from both sides of the equation gives $2x^2 - 4x - t = 0$. In this form, $a = 2$, $b = -4$, and $c = -t$. The values of t for which the equation has no real solutions are the same values of t for which the discriminant of this equation is a negative value. The discriminant is equal to $(-4)^2 - 4(2)(-t)$; therefore, $(-4)^2 - 4(2)(-t) < 0$. Simplifying the left side of the inequality gives $16 + 8t < 0$. Subtracting 16 from both sides of the inequality and then dividing both sides by 8 gives $t < -2$. Of the values given in the options, -3 is the only value that is less than -2 . Therefore, choice A must be the correct answer.

Choices B, C, and D are incorrect and may result from a misconception about how to use the discriminant to determine the number of solutions of a quadratic equation in one variable.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A certain bowl was sold online for prices varying between \$7 and \$13. The number of bowls sold can be modeled by a quadratic function based on the price, where 5 bowls were sold for \$12 each and the maximum of 9 bowls were sold for \$10 each. According to the model, what is the number of bowls that were sold for \$8 each?

Correct Answer: 5

Rationale

The correct answer is 5. It's given that the number of bowls sold can be modeled by a quadratic function based on the price, where a maximum of 9 bowls were sold for \$10 each and 5 bowls were sold for \$12 each. This quadratic function can be defined by an equation of the form $N(p) = a(p - h)^2 + k$, where $N(p)$ is the number of bowls sold when the price is \$ p each, and a , h , and k are constants such that the function reaches its maximum value, k , when $p = h$. Since the model indicates the maximum number of bowls sold is 9 when the price is \$10 each, $N(p)$ reaches its maximum value, 9, when $p = 10$. Therefore, $k = 9$ and $h = 10$. Substituting 10 for h and 9 for k in the function $N(p) = a(p - h)^2 + k$ yields $N(p) = a(p - 10)^2 + 9$. Since the model indicates the number of bowls sold is 5 when the price is \$12 each, the value of $N(p)$ is 5 when $p = 12$. Substituting 12 for p and 5 for $N(p)$ in the equation $N(p) = a(p - 10)^2 + 9$ yields $5 = a(12 - 10)^2 + 9$, or $5 = 4a + 9$. Subtracting 9 from both sides of this equation yields $-4 = 4a$. Dividing both sides of this equation by 4 yields $-1 = a$. Therefore, the model can be represented by the equation $N(p) = -(p - 10)^2 + 9$. Substituting 8 for p in this equation yields $N(8) = -(8 - 10)^2 + 9$, or $N(8) = -4 + 9$, or $N(8) = 5$. Therefore, according to the model, the number of bowls sold for \$8 each was 5.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$-2x^2 + 20x + c = 0$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

Answer

- A. -68
- B. -50
- C. -32
- D. 0

Correct Answer: B

Rationale

Choice B is correct. It's given that the equation $-2x^2 + 20x + c = 0$, where c is a constant, has exactly one solution. A quadratic equation of the form $ax^2 + bx + c = 0$ has exactly one solution if and only if its discriminant, $b^2 - 4ac$, is equal to zero. It follows that for the given equation, $a = -2$ and $b = 20$. Substituting -2 for a and 20 for b in $b^2 - 4ac$ yields $20^2 - 4(-2)(c)$, or $400 + 8c$. Since the discriminant must equal zero, it follows that $400 + 8c = 0$. Subtracting 400 from both sides of this equation yields $8c = -400$. Dividing each side of this equation by 8 yields $c = -50$. Therefore, the value of c is -50 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 8462b105

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f gives the product of a number, x , and a number that is **91** more than x . Which equation defines f ?

Answer

- A. $f(x) = x^2 + x + 91$
- B. $f(x) = x^2 + 91$
- C. $f(x) = x^2 + 91x$
- D. $f(x) = x^2 + 91x + 91$

Correct Answer: C

Rationale

Choice C is correct. It’s given that the function f gives the product of a number, x , and a number that is **91** more than x . A number that is **91** more than x can be represented by the expression $x + 91$. Therefore, f can be defined by the equation $f(x) = x(x + 91)$, or $f(x) = x^2 + 91x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A model estimates that at the end of each year from **2015** to **2020**, the number of squirrels in a population was **150%** more than the number of squirrels in the population at the end of the previous year. The model estimates that at the end of **2016**, there were **180** squirrels in the population. Which of the following equations represents this model, where n is the estimated number of squirrels in the population t years after the end of **2015** and $t \leq 5$?

Answer

- A. $n = 72(1.5)^t$
- B. $n = 72(2.5)^t$
- C. $n = 180(1.5)^t$
- D. $n = 180(2.5)^t$

Correct Answer: B

Rationale

Choice B is correct. Since the model estimates that the number of squirrels in the population increased by a fixed percentage, **150%**, each year, the model can be represented by an exponential equation of the form $n = a\left(1 + \frac{p}{100}\right)^t$, where a is the estimated number of squirrels in the population at the end of **2015**, and the model estimates that at the end of each year, the number is $p\%$ more than the number at the end of the previous year. Since the model estimates that at the end of each year, the number was **150%** more than the number at the end of the previous year, $p = 150$. Substituting **150** for p in the equation $n = a\left(1 + \frac{p}{100}\right)^t$ yields $n = a\left(1 + \frac{150}{100}\right)^t$, which is equivalent to $n = a(1 + 1.5)^t$, or $n = a(2.5)^t$. It's given that the estimated number of squirrels at the end of **2016** was **180**. This means that when $t = 1$, $n = 180$. Substituting **1** for t and **180** for n in the equation $n = a(2.5)^t$ yields $180 = a(2.5)^1$, or $180 = 2.5a$. Dividing each side of this equation by **2.5** yields $72 = a$. Substituting **72** for a in the equation $n = a(2.5)^t$ yields $n = 72(2.5)^t$.

Choice A is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was **150%** of, not **150%** more than, the estimated number at the end of the previous year.

Choice C is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was **150%** of, not **150%** more than, the estimated number at the end of the previous year, and the estimated number of squirrels at the end of **2015**, not the end of **2016**, was **180**.

Choice D is incorrect. This equation represents a model where the estimated number of squirrels at the end of **2015**, not the end of **2016**, was **180**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$0.36x^2 + 0.63x + 1.17$

The given expression can be rewritten as $a(4x^2 + 7x + 13)$, where a is a constant. What is the value of a ?

Correct Answer: .09, 9/100

Rationale

The correct answer is .09. It's given that the expression $0.36x^2 + 0.63x + 1.17$ can be rewritten as $a(4x^2 + 7x + 13)$. Applying the distributive property to the expression $a(4x^2 + 7x + 13)$ yields $4ax^2 + 7ax + 13a$. Therefore, $0.36x^2 + 0.63x + 1.17$ can be rewritten as $4ax^2 + 7ax + 13a$. It follows that in the expressions $0.36x^2 + 0.63x + 1.17$ and $4ax^2 + 7ax + 13a$, the coefficients of x^2 are equivalent, the coefficients of x are equivalent, and the constant terms are equivalent. Therefore, $0.36 = 4a$, $0.63 = 7a$, and $1.17 = 13a$. Solving any of these equations for a yields the value of a . Dividing both sides of the equation $0.36 = 4a$ by 4 yields $0.09 = a$. Therefore, the value of a is 0.09. Note that .09 and 9/100 are examples of ways to enter a correct answer.

Question ID: c81b6c57

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

In the expression $3(2x^2 + px + 8) - 16x(p + 4)$, p is a constant. This expression is equivalent to the expression $6x^2 - 155x + 24$. What is the value of p ?

Answer

- A. -3
- B. 7
- C. 13
- D. 155

Correct Answer: B

Rationale

Choice B is correct. Using the distributive property, the first given expression can be rewritten as $6x^2 + 3px + 24 - 16px - 64x + 24$, and then rewritten as $6x^2 + (3p - 16p - 64)x + 24$. Since the expression $6x^2 + (3p - 16p - 64)x + 24$ is equivalent to $6x^2 - 155x + 24$, the coefficients of the x terms from each expression are equivalent to each other; thus $3p - 16p - 64 = -155$. Combining like terms gives $-13p - 64 = -155$. Adding 64 to both sides of the equation gives $-13p = -91$. Dividing both sides of the equation by -13 yields $p = 7$.

Choice A is incorrect. If $p = -3$, then the first expression would be equivalent to $6x^2 - 25x + 24$. Choice C is incorrect. If $p = 13$, then the first expression would be equivalent to $6x^2 - 233x + 24$. Choice D is incorrect. If $p = 155$, then the first expression would be equivalent to $6x^2 - 2,079x + 24$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(t) = 55t - 2t^2$

The function f is defined by the given equation. The function g is defined by $g(t) = f(t) + 3$. Which expression represents the maximum value of $g(t)$?

Answer

- A. $3 + \left(\frac{55}{2}\right)^2$
- B. $3 + 2\left(\frac{55}{4}\right)^2$
- C. $3 - 2\left(\frac{55}{4}\right)^2$
- D. $3 - \left(\frac{55}{2}\right)^2$

Correct Answer: B

Rationale

Choice B is correct. It's given that function g is defined by $g(t) = f(t) + 3$ and that $f(t) = 55t - 2t^2$. Substituting $55t - 2t^2$ for $f(t)$ in $g(t) = f(t) + 3$ yields $g(t) = 55t - 2t^2 + 3$, or $g(t) = -2t^2 + 55t + 3$. The maximum value of $g(t)$ can be found by completing the square to rewrite the equation defining g in the form $g(t) = a(t - h)^2 + k$, where the maximum value of the function is k , which occurs when $t = h$, and a is a negative constant. The equation $g(t) = -2t^2 + 55t + 3$ is equivalent to $g(t) = -2\left(t^2 - \frac{55}{2}t\right) + 3$, which can be rewritten as $g(t) = -2\left(t^2 - \frac{55}{2}t + \left(\frac{55}{4}\right)^2\right) + 3 + 2\left(\frac{55}{4}\right)^2$, or $g(t) = -2\left(t - \frac{55}{4}\right)^2 + 3 + 2\left(\frac{55}{4}\right)^2$. This equation is in the form $g(t) = a(t - h)^2 + k$, where $a = -2$, $h = \frac{55}{4}$, and $k = 3 + 2\left(\frac{55}{4}\right)^2$. Thus, the maximum value of $g(t)$ is $3 + 2\left(\frac{55}{4}\right)^2$.

Alternate approach: Since the function f is a quadratic function, the maximum value of $f(t)$ occurs at the value of t that's halfway between the two zeros of the function. The zeros of function f can be found by substituting 0 for $f(t)$ in the equation defining f , which yields $0 = 55t - 2t^2$. This equation can be rewritten as $0 = t(55 - 2t)$. By the zero product property, it follows that $t = 0$ or $55 - 2t = 0$. Subtracting 55 from each side of the equation $55 - 2t = 0$ yields $-2t = -55$. Dividing each side of this equation by -2 yields $t = \frac{55}{2}$. Therefore, the zeros of function f are 0 and $\frac{55}{2}$. The value that's halfway between 0 and $\frac{55}{2}$ can be found by calculating the average of 0 and $\frac{55}{2}$, which is $\frac{0 + \frac{55}{2}}{2}$, or $\frac{55}{4}$. It follows that the maximum of function f occurs when $t = \frac{55}{4}$. Substituting $\frac{55}{4}$ for t in the equation defining function f yields $f\left(\frac{55}{4}\right) = 55\left(\frac{55}{4}\right) - 2\left(\frac{55}{4}\right)^2$, which is equivalent to $f\left(\frac{55}{4}\right) = \frac{55^2}{4} - 2\left(\frac{55^2}{4^2}\right)$. Multiplying $\frac{55^2}{4}$ by $\frac{4}{4}$ in this equation to get a common denominator yields $f\left(\frac{55}{4}\right) = 4\left(\frac{55^2}{4^2}\right) - 2\left(\frac{55^2}{4^2}\right)$, or $f\left(\frac{55}{4}\right) = 2\left(\frac{55^2}{4^2}\right)$, which is equivalent to $f\left(\frac{55}{4}\right) = 2\left(\frac{55}{4}\right)^2$. Thus, the maximum value of $f(t)$ is $2\left(\frac{55}{4}\right)^2$. Since the equation defining $g(t)$ is $g(t) = f(t) + 3$, the maximum value of $g(t)$ is 3 greater than the maximum value of $f(t)$. It follows that the maximum value of $g(t)$ is $3 + 2\left(\frac{55}{4}\right)^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$\sqrt{2x+6} + 4 = x + 3$

What is the solution set of the equation above?

Answer

- A. $\{-1\}$
- B. $\{5\}$
- C. $\{-1, 5\}$
- D. $\{0, -1, 5\}$

Correct Answer: B

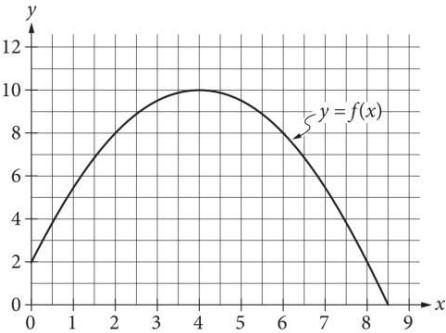
Rationale

Choice B is correct. Subtracting 4 from both sides of $\sqrt{2x+6} + 4 = x + 3$ isolates the radical expression on the left side of the equation as follows: $\sqrt{2x+6} = x - 1$. Squaring both sides of $\sqrt{2x+6} = x - 1$ yields $2x + 6 = x^2 - 2x + 1$. This equation can be rewritten as a quadratic equation in standard form: $x^2 - 4x - 5 = 0$. One way to solve this quadratic equation is to factor the expression $x^2 - 4x - 5$ by identifying two numbers with a sum of -4 and a product of -5 . These numbers are -5 and 1. So the quadratic equation can be factored as $(x - 5)(x + 1) = 0$. It follows that 5 and -1 are the solutions to the quadratic equation. However, the solutions must be verified by checking whether 5 and -1 satisfy the original equation, $\sqrt{2x+6} + 4 = x + 3$. When $x = -1$, the original equation gives $\sqrt{2(-1)+6} + 4 = (-1) + 3$, or $6 = 2$, which is false. Therefore, -1 does not satisfy the original equation. When $x = 5$, the original equation gives $\sqrt{2(5)+6} + 4 = 5 + 3$, or $8 = 8$, which is true. Therefore, $x = 5$ is the only solution to the original equation, and so the solution set is $\{5\}$.

Choices A, C, and D are incorrect because each of these sets contains at least one value that results in a false statement when substituted into the given equation. For instance, in choice D, when 0 is substituted for x into the given equation, the result is $\sqrt{2(0)+6} + 4 = (0) + 3$, or $\sqrt{6} + 4 = 3$. This is not a true statement, so 0 is not a solution to the given equation.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question



The graph of the function f , defined by $f(x) = -\frac{1}{2}(x-4)^2 + 10$, is shown in the xy -plane above. If the function g (not shown) is defined by $g(x) = -x + 10$, what is one possible value of a such that $f(a) = g(a)$?

Rationale

The correct answer is either 2 or 8. Substituting $x = a$ in the definitions for f and g gives $f(a) = -\frac{1}{2}(a-4)^2 + 10$ and $g(a) = -a + 10$, respectively. If $f(a) = g(a)$, then $-\frac{1}{2}(a-4)^2 + 10 = -a + 10$. Subtracting 10 from both sides of this equation gives $-\frac{1}{2}(a-4)^2 = -a$. Multiplying both sides by -2 gives $(a-4)^2 = 2a$. Expanding $(a-4)^2$ gives $a^2 - 8a + 16 = 2a$. Combining the like terms on one side of the equation gives $a^2 - 10a + 16 = 0$. One way to solve this equation is to factor $a^2 - 10a + 16$ by identifying two numbers with a sum of -10 and a product of 16. These numbers are -2 and -8 , so the quadratic equation can be factored as $(a-2)(a-8) = 0$. Therefore, the possible values of a are either 2 or 8. Note that 2 and 8 are examples of ways to enter a correct answer.

Alternate approach: Graphically, the condition $f(a) = g(a)$ implies the graphs of the functions $y = f(x)$ and $y = g(x)$ intersect at $x = a$. The graph $y = f(x)$ is given, and the graph of $y = g(x)$ may be sketched as a line with y -intercept 10 and a slope of -1 (taking care to note the different scales on each axis). These two graphs intersect at $x = 2$ and $x = 8$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The population P of a certain city y years after the last census is modeled by the equation below, where r is a constant and P_0 is the population when $y = 0$.

$$P = P_0(1 + r)^y$$

If during this time the population of the city decreases by a fixed percent each year, which of the following must be true?

Answer

- A. $r < -1$
- B. $-1 < r < 0$
- C. $0 < r < 1$
- D. $r > 1$

Correct Answer: B

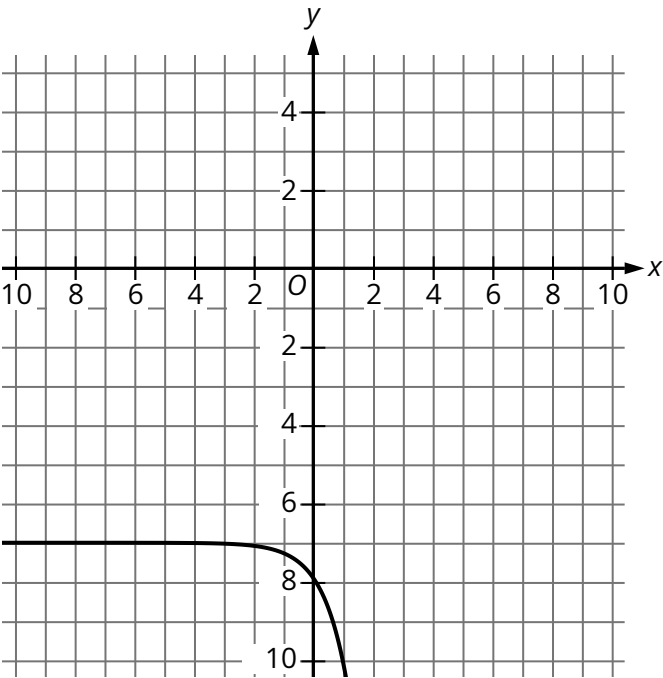
Rationale

Choice B is correct. The term $(1 + r)$ represents a percent change. Since the population is decreasing, the percent change must be between 0% and 100%. When the percent change is expressed as a decimal rather than as a percent, the percentage change must be between 0 and 1. Because $(1 + r)$ represents percent change, this can be expressed as $0 < 1 + r < 1$. Subtracting 1 from all three terms of this compound inequality results in $-1 < r < 0$.

Choice A is incorrect. If $r < -1$, then after 1 year, the population P would be a negative value, which is not possible. Choices C and D are incorrect. For any value of $r > 0$, $1 + r > 1$, and the exponential function models growth for positive values of the exponent. This contradicts the given information that the population is decreasing.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question



The graph of $y = f(x)$ is shown, where $f(x) = ab^x + c$, and a , b , and c are constants. For how many values of x does $f(x) = 0$?

Answer

- A. Three
- B. Two
- C. One
- D. Zero

Correct Answer: D

Rationale

Choice D is correct. Each point (x, y) on the graph of $y = f(x)$ in the xy -plane gives a value of x and its corresponding value of $f(x)$, or y . For any value of x for which $f(x) = 0$, there is a corresponding point on the graph of $y = f(x)$ with a y -coordinate of 0 . A point with a y -coordinate of 0 is a point where the graph intersects the x -axis. It's given that $f(x) = ab^x + c$, where a , b , and c are constants. In the xy -plane, the graph of an equation of this form will lie entirely either above or below the horizontal line defined by $y = c$. The part of the graph of $y = f(x)$ shown lies entirely below the horizontal line defined by $y = -7$, and thus the entire graph of $y = f(x)$ must lie below the line defined by $y = -7$. It follows that the graph of $y = f(x)$ will never intersect the x -axis. Therefore, there are zero values of x for which $f(x) = 0$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question ID: 3d12b1e0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$-16x^2 - 8x + c = 0$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

Correct Answer: -1

Rationale

The correct answer is -1 . A quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has exactly one solution when its discriminant, $b^2 - 4ac$, is equal to 0 . In the given equation, $-16x^2 - 8x + c = 0$, $a = -16$ and $b = -8$. Substituting -16 for a and -8 for b in $b^2 - 4ac$ yields $(-8)^2 - 4(-16)(c)$, or $64 + 64c$. Since the given equation has exactly one solution, $64 + 64c = 0$. Subtracting 64 from both sides of this equation yields $64c = -64$. Dividing both sides of this equation by 64 yields $c = -1$. Therefore, the value of c is -1 .

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The expression $\frac{x^{-2}y^{\frac{1}{2}}}{x^{\frac{1}{3}}y^{-1}}$, where $x > 1$ and $y > 1$, is equivalent to which of the following?

Answer

- A. $\frac{\sqrt{y}}{\sqrt[3]{x^2}}$
- B. $\frac{y\sqrt{y}}{\sqrt[3]{x^2}}$
- C. $\frac{y\sqrt{y}}{x\sqrt{x}}$
- D. $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$

Correct Answer: D

Rationale

Choice D is correct. For $x > 1$ and $y > 1$, $x^{\frac{1}{3}}$ and $y^{\frac{1}{2}}$ are equivalent to $\sqrt[3]{x}$ and $\sqrt{y}$, respectively. Also, x^{-2} and y^{-1} are equivalent to $\frac{1}{x^2}$ and $\frac{1}{y}$, respectively. Therefore, the given expression can be rewritten as $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$.

Choices A, B, and C are incorrect because these choices are not equivalent to the given expression for $x > 1$ and $y > 1$.

For example, for $x = 2$ and $y = 2$, the value of the given expression is $2^{-\frac{5}{6}}$; the values of the choices, however, are $2^{-\frac{1}{3}}$, $2^{\frac{5}{6}}$, and 1, respectively.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$\frac{1}{4xy} + xyz = \frac{1}{3yz}$

In the given equation, x , y , and z are positive numbers. Which expression is equivalent to y ?

Answer

- A. $\frac{4x-3z}{12x^2z^2}$
- B. $\sqrt{\frac{4x-3z}{12x^2z^2}}$
- C. $\frac{1}{3xz^2-4x^2z}$
- D. $\sqrt{\frac{1}{3xz^2-4x^2z}}$

Correct Answer: B

Rationale

Choice B is correct. It's given that in the equation $\frac{1}{4xy} + xyz = \frac{1}{3yz}$, x , y , and z are positive numbers. Multiplying both sides of this equation by the least common denominator of $12xyz$ yields $12xyz\left(\frac{1}{4xy} + xyz\right) = 12xyz\left(\frac{1}{3yz}\right)$, which gives $\frac{12xyz}{4xy} + 12x^2y^2z^2 = \frac{12xyz}{3yz}$, or $3z + 12x^2y^2z^2 = 4x$. Subtracting $3z$ from both sides of this equation yields $12x^2y^2z^2 = 4x - 3z$. Dividing both sides of this equation by $12x^2z^2$ yields $y^2 = \frac{4x-3z}{12x^2z^2}$. Taking the square root of both sides of this equation yields $y = \pm\sqrt{\frac{4x-3z}{12x^2z^2}}$. Since y is a positive number, the expression $\sqrt{\frac{4x-3z}{12x^2z^2}}$ is equivalent to y .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 71014fb1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$(x - 1)^2 = -4$

How many distinct real solutions does the given equation have?

Answer

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Correct Answer: D

Rationale

Choice D is correct. Any quantity that is positive or negative in value has a positive value when squared. Therefore, the left-hand side of the given equation is either positive or zero for any value of x . Since the right-hand side of the given equation is negative, there is no value of x for which the given equation is true. Thus, the number of distinct real solutions for the given equation is zero.

Choices A, B, and C are incorrect and may result from conceptual errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$f(x) = x^3 - 9x$
 $g(x) = x^2 - 2x - 3$

Which of the following expressions is equivalent to $\frac{f(x)}{g(x)}$, for $x > 3$?

Answer

- A. $\frac{1}{x+1}$
- B. $\frac{x+3}{x+1}$
- C. $\frac{x(x-3)}{x+1}$
- D. $\frac{x(x+3)}{x+1}$

Correct Answer: D

Rationale

Choice D is correct. Since $x^3 - 9x = x(x+3)(x-3)$ and $x^2 - 2x - 3 = (x+1)(x-3)$, the fraction $\frac{f(x)}{g(x)}$ can be written as $\frac{x(x+3)(x-3)}{(x+1)(x-3)}$. It is given that $x > 3$, so the common factor $x - 3$ is not equal to 0. Therefore, the fraction can be further simplified to $\frac{x(x+3)}{x+1}$.

Choice A is incorrect. The expression $\frac{1}{x+1}$ is not equivalent to $\frac{f(x)}{g(x)}$ because at $x = 0$, $\frac{1}{x+1}$ has a value of 1 and $\frac{f(x)}{g(x)}$ has a value of 0.

Choice B is incorrect and results from omitting the factor x in the factorization of $f(x)$. Choice C is incorrect and may result from incorrectly factoring $g(x)$ as $(x+1)(x+3)$ instead of $(x+1)(x-3)$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The expression $\frac{1}{3}x^2 - 2$ can be rewritten as $\frac{1}{3}(x - k)(x + k)$, where k is a positive constant. What is the value of k ?

Answer

- A. 2
- B. 6
- C. $\sqrt{2}$
- D. $\sqrt{6}$

Correct Answer: D

Rationale

Choice D is correct. Factoring out the coefficient $\frac{1}{3}$, the given expression can be rewritten as $\frac{1}{3}(x^2 - 6)$. The expression $x^2 - 6$ can be approached as a difference of squares and rewritten as $(x - \sqrt{6})(x + \sqrt{6})$. Therefore, k must be $\sqrt{6}$.

Choice A is incorrect. If k were 2, then the expression given would be rewritten as $\frac{1}{3}(x - 2)(x + 2)$, which is equivalent to $\frac{1}{3}x^2 - \frac{4}{3}$, not $\frac{1}{3}x^2 - 2$.

Choice B is incorrect. This may result from incorrectly factoring the expression and finding $(x - 6)(x + 6)$ as the factored form of the expression. Choice C is incorrect. This may result from incorrectly distributing the $\frac{1}{3}$ and rewriting the expression as $\frac{1}{3}(x^2 - 2)$.

Question ID: 5acbd30

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question
The function f is defined by $f(x) = 56(0.19)^x$. For any positive integer n , the value of $f(n)$ is $p\%$ less than the value of $f(n - 1)$. What is the value of p ?

- Answer**
- A. 19
 - B. 44
 - C. 56
 - D. 81

Correct Answer: D

Rationale
Choice D is correct. For a function of the form $f(x) = a(b)^x$, where a and b are constants and $b < 1$, the value of $f(x)$ decreases by $100(1 - b)\%$ for every increase of x by 1. In the given function, $a = 56$ and $b = 0.19$. Therefore, for the given function, the value of $f(x)$ decreases by $100(1 - 0.19)\%$, or 81% , for every increase of x by 1. It's given that the value of $f(n)$ is $p\%$ less than the value of $f(n - 1)$. The increase from $x = n - 1$ to $x = n$ can be expressed as $n - (n - 1)$, which is equivalent to $n - n + 1$, or 1. Since there is an increase by 1 from $n - 1$ to n , it follows that the value of $f(n)$ is 81% less than the value of $f(n - 1)$. Thus, the value of p is 81.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$y = x^2 + 2x + 1$
 $x + y + 1 = 0$

If (x_1, y_1) and (x_2, y_2) are the two solutions to the system of equations above, what is the value of $y_1 + y_2$?

Answer

- A. -3
- B. -2
- C. -1
- D. 1

Correct Answer: D

Rationale

Choice D is correct. The system of equations can be solved using the substitution method. Solving the second equation for y gives $y = -x - 1$. Substituting the expression $-x - 1$ for y into the first equation gives $-x - 1 = x^2 + 2x + 1$. Adding $x + 1$ to both sides of the equation yields $x^2 + 3x + 2 = 0$. The left-hand side of the equation can be factored by finding two numbers whose sum is 3 and whose product is 2, which gives $(x + 2)(x + 1) = 0$. Setting each factor equal to 0 yields $x + 2 = 0$ and $x + 1 = 0$, and solving for x yields $x = -2$ or $x = -1$. These values of x can be substituted for x in the equation $y = -x - 1$ to find the corresponding y-values: $y = -(-2) - 1 = 2 - 1 = 1$ and $y = -(-1) - 1 = 1 - 1 = 0$. It follows that $(-2, 1)$ and $(-1, 0)$ are the solutions to the given system of equations. Therefore, $(x_1, y_1) = (-2, 1)$, $(x_2, y_2) = (-1, 0)$, and $y_1 + y_2 = 1 + 0 = 1$.

Choice A is incorrect. The solutions to the system of equations are $(x_1, y_1) = (-2, 1)$ and $(x_2, y_2) = (-1, 0)$. Therefore, -3 is the sum of the x-coordinates of the solutions, not the sum of the y-coordinates of the solutions. Choices B and C are incorrect and may be the result of computation or substitution errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$h(t) = -16t^2 + b$$

The function h estimates a ball's height, in feet, above a floor t seconds after the ball is released, where b is a constant. The function estimates that the ball is **19.36** feet above the floor when it is released at $t = 0$. How many seconds after being released does the function estimate the ball will reach the floor?

Correct Answer: 1.1, 11/10

Rationale

The correct answer is **1.1**. It's given that $h(t) = -16t^2 + b$ estimates a ball's height, in feet, above a floor t seconds after the ball is released, where b is a constant. It's also given that the function estimates that the ball is **19.36** feet above the floor when it is released at $t = 0$. Substituting **19.36** for $h(t)$ and **0** for t in the given function yields $19.36 = -16(0)^2 + b$, or $19.36 = b$. Substituting **19.36** for b in function h yields $h(t) = -16t^2 + 19.36$. When the ball reaches the floor, its height is **0** feet above the floor. Substituting **0** for $h(t)$ yields $0 = -16t^2 + 19.36$. Adding $16t^2$ to both sides of this equation yields $16t^2 = 19.36$. Dividing both sides of this equation by **16** yields $t^2 = 1.21$. Taking the positive square root of both sides of this equation yields $t = 1.1$. Therefore, the function estimates that the ball will reach the floor **1.1** seconds after being released. Note that 1.1 and 11/10 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

If $u - 3 = \frac{6}{t - 2}$, what is t in terms of u ?

Answer

- A. $t = \frac{1}{u}$
- B. $t = \frac{2u + 9}{u}$
- C. $t = \frac{1}{u - 3}$
- D. $t = \frac{2u}{u - 3}$

Correct Answer: D

Rationale

Choice D is correct. Multiplying both sides of the given equation by $t - 2$ yields $(t - 2)(u - 3) = 6$. Dividing both sides of this equation by $u - 3$ yields $t - 2 = \frac{6}{u - 3}$. Adding 2 to both sides of this equation yields $t = \frac{6}{u - 3} + 2$, which can be rewritten as $t = \frac{6}{u - 3} + \frac{2(u - 3)}{u - 3}$. Since the fractions on the right-hand side of this equation have a common denominator, adding the fractions yields $t = \frac{6 + 2(u - 3)}{u - 3}$. Applying the distributive property to the numerator on the right-hand side of this equation yields $t = \frac{6 + 2u - 6}{u - 3}$, which is equivalent to $t = \frac{2u}{u - 3}$.

Choices A, B, and C are incorrect and may result from various misconceptions or miscalculations.

Question ID: 7bf77c19

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The functions p and r are defined by the equations shown, where a and b are integer constants such that $a < b$ and $b < 0$. If $y = p(x)$ and $y = r(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

- I. $p(x) = a(4.1)^x + b$
- II. $r(x) = a(4.1)^{x+b}$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Correct Answer: D

Rationale

Choice D is correct. It's given that a and b are integer constants such that $a < b$ and $b < 0$. The y -coordinate of the y -intercept of the graph of a function is the value of the function when $x = 0$. Equation I is $p(x) = a(4.1)^x + b$. Substituting 0 for x in this equation yields $p(0) = a(4.1)^0 + b$, which can be rewritten as $p(0) = a(1) + b$, or $p(0) = a + b$. Therefore, the y -coordinate of the y -intercept of the graph of $y = p(x)$ is $a + b$. Equation I has a coefficient of a and a constant b . Thus, the value of the y -coordinate of the y -intercept, $a + b$, is not displayed as a constant or coefficient in equation I. Equation II is $r(x) = a(4.1)^{x+b}$. Similarly, substituting 0 for x in this equation yields $r(0) = a(4.1)^{0+b}$, which can be rewritten as $r(0) = a(4.1)^b$. Since $b < 0$, the value of $(4.1)^b$ is not equal to 1. Thus, the value of $a(4.1)^b$ is not equal to a . The value of $a(4.1)^b$ is also not equal to b , since there is no pair of integers a and b such that $a(4.1)^b = b$. Thus, the value of the y -coordinate of the y -intercept for this equation is $a(4.1)^b$, which is not displayed in equation II as a constant or coefficient. The coefficient in equation II is a , and the only constant, b , appears in the exponent. Therefore, neither equation I nor equation II displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$8x + y = -11$

$2x^2 = y + 341$

The graphs of the equations in the given system of equations intersect at the point (x, y) in the xy -plane. What is a possible value of x ?

Answer

- A. -15
- B. -11
- C. 2
- D. 8

Correct Answer: A

Rationale

Choice A is correct. It's given that the graphs of the equations in the given system of equations intersect at the point (x, y) . Therefore, this intersection point is a solution to the given system. The solution can be found by isolating y in each equation. The given equation $8x + y = -11$ can be rewritten to isolate y by subtracting $8x$ from both sides of the equation, which gives $y = -8x - 11$. The given equation $2x^2 = y + 341$ can be rewritten to isolate y by subtracting 341 from both sides of the equation, which gives $2x^2 - 341 = y$. With each equation solved for y , the value of y from one equation can be substituted into the other, which gives $2x^2 - 341 = -8x - 11$. Adding $8x$ and 11 to both sides of this equation results in $2x^2 + 8x - 330 = 0$. Dividing both sides of this equation by 2 results in $x^2 + 4x - 165 = 0$. This equation can be rewritten by factoring the left-hand side, which yields $(x + 15)(x - 11) = 0$. By the zero-product property, if $(x + 15)(x - 11) = 0$, then $(x + 15) = 0$, or $(x - 11) = 0$. It follows that $x = -15$, or $x = 11$. Since only -15 is given as a choice, a possible value of x is -15 .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

x	p(x)
-2	5
-1	0
0	-3
1	-1
2	0

The table above gives selected values of a polynomial function p. Based on the values in the table, which of the following must be a factor of p ?

- Answer
- A. $(x - 3)$
- B. $(x + 3)$
- C. $(x - 1)(x + 2)$
- D. $(x + 1)(x - 2)$

Correct Answer: D

Rationale

Choice D is correct. According to the table, when x is -1 or 2 , $p(x) = 0$. Therefore, two x-intercepts of the graph of p are $(-1, 0)$ and $(2, 0)$. Since $(-1, 0)$ and $(2, 0)$ are x-intercepts, it follows that $(x + 1)$ and $(x - 2)$ are factors of the polynomial equation. This is because when $x = -1$, the value of $x + 1$ is 0. Similarly, when $x = 2$, the value of $x - 2$ is 0. Therefore, the product $(x + 1)(x - 2)$ is a factor of the polynomial function p.

Choice A is incorrect. The factor $x - 3$ corresponds to an x-intercept of $(3, 0)$, which isn't present in the table. Choice B is incorrect. The factor $x + 3$ corresponds to an x-intercept of $(-3, 0)$, which isn't present in the table. Choice C is incorrect. The factors $x - 1$ and $x + 2$ correspond to x-intercepts $(1, 0)$ and $(-2, 0)$, respectively, which aren't present in the table.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

When the quadratic function f is graphed in the xy -plane, where $y = f(x)$, its vertex is $(-3, 6)$. One of the x -intercepts of this graph is $(-\frac{17}{4}, 0)$. What is the other x -intercept of the graph?

Answer

- A. $(-\frac{29}{4}, 0)$
- B. $(-\frac{7}{4}, 0)$
- C. $(\frac{5}{4}, 0)$
- D. $(\frac{17}{4}, 0)$

Correct Answer: B

Rationale

Choice B is correct. Since the line of symmetry for the graph of a quadratic function contains the vertex of the graph, the x -coordinate of the vertex of the graph of $y = f(x)$ is the x -coordinate of the midpoint of its two x -intercepts. The midpoint of two points with x -coordinates x_1 and x_2 has x -coordinate x_m , where $x_m = \frac{x_1+x_2}{2}$. It's given that the vertex is $(-3, 6)$ and one of the x -intercepts is $(-\frac{17}{4}, 0)$. Substituting -3 for x_m and $-\frac{17}{4}$ for x_1 in the equation $x_m = \frac{x_1+x_2}{2}$ yields $-3 = \frac{-\frac{17}{4}+x_2}{2}$. Multiplying each side of this equation by 2 yields $-6 = -\frac{17}{4} + x_2$. Adding $\frac{17}{4}$ to each side of this equation yields $-\frac{7}{4} = x_2$. Therefore, the other x -intercept is $(-\frac{7}{4}, 0)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 668f1863

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question
Function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. The graph of $y = f(x)$ in the xy -plane has a vertex at $(r, -64)$. What is the value of r ?

Correct Answer: -12

Rationale
The correct answer is -12 . It's given that function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. It follows that the graph of $y = f(x)$ in the xy -plane passes through the points $(-20, 0)$ and $(-4, 0)$. When the graph of a quadratic function contains two points $(a, 0)$ and $(b, 0)$, the x -coordinate of the vertex of the graph is the average of a and b . Therefore, the x -coordinate of the vertex of the graph of $y = f(x)$ is $\frac{-20+(-4)}{2}$, or -12 . It's given that the graph of $y = f(x)$ in the xy -plane has a vertex at $(r, -64)$. It follows that the value of r is -12 .

Question ID: 70753f99

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is defined by $f(x) = (x + 3)(x + 1)$. The graph of f in the xy -plane is a parabola. Which of the following intervals contains the x -coordinate of the vertex of the graph of f ?

Answer

- A. $-4 < x < -3$
- B. $-3 < x < 1$
- C. $1 < x < 3$
- D. $3 < x < 4$

Correct Answer: B

Rationale

Choice B is correct. The graph of a quadratic function in the xy -plane is a parabola. The axis of symmetry of the parabola passes through the vertex of the parabola. Therefore, the vertex of the parabola and the midpoint of the segment between the two x -intercepts of the graph have the same x -coordinate. Since $f(-3) = f(-1) = 0$, the x -coordinate of the vertex is $\frac{(-3) + (-1)}{2} = -2$. Of the shown intervals, only the interval in choice B contains -2 . Choices A, C, and D are incorrect and may result from either calculation errors or misidentification of the graph's x -intercepts.

Question ID: 58dcc59f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A landscaper is designing a rectangular garden. The length of the garden is to be 5 feet longer than the width. If the area of the garden will be 104 square feet, what will be the length, in feet, of the garden?

Rationale

The correct answer is 13. Let w represent the width of the rectangular garden, in feet. Since the length of the garden will be 5 feet longer than the width of the garden, the length of the garden will be $w + 5$ feet. Thus the area of the garden will be $w(w + 5)$. It is also given that the area of the garden will be 104 square feet. Therefore, $w(w + 5) = 104$, which is equivalent to $w^2 + 5w - 104 = 0$. Factoring this equation results in $(w + 13)(w - 8) = 0$. Therefore, $w = 8$ and $w = -13$. Because width cannot be negative, the width of the garden must be 8 feet. This means the length of the garden must be $8 + 5 = 13$ feet.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

In the xy -plane, the graph of $y = x^2 - 9$ intersects line p at $(1, a)$ and $(5, b)$, where a and b are constants. What is the slope of line p ?

Answer

- A. 6
- B. 2
- C. -2
- D. -6

Correct Answer: A

Rationale

Choice A is correct. It's given that the graph of $y = x^2 - 9$ and line p intersect at $(1, a)$ and $(5, b)$. Therefore, the value of y when $x = 1$ is the value of a , and the value of y when $x = 5$ is the value of b . Substituting 1 for x in the given equation yields $y = (1)^2 - 9$, or $y = -8$. Similarly, substituting 5 for x in the given equation yields $y = (5)^2 - 9$, or $y = 16$. Therefore, the intersection points are $(1, -8)$ and $(5, 16)$. The slope of line p is the ratio of the change in y to the change in x between these two points: $\frac{16 - (-8)}{5 - 1} = \frac{24}{4}$, or 6.

Choices B, C, and D are incorrect and may result from conceptual or calculation errors in determining the values of a , b , or the slope of line p .

Question ID: 84dd43f8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

For the function f , $f(0) = 86$, and for each increase in x by 1, the value of $f(x)$ decreases by 80%. What is the value of $f(2)$?

Correct Answer: 3.44, 86/25

Rationale

The correct answer is **3.44**. It's given that $f(0) = 86$ and that for each increase in x by 1, the value of $f(x)$ decreases by 80%. Because the output of the function decreases by a constant percentage for each 1-unit increase in the value of x , this relationship can be represented by an exponential function of the form $f(x) = a(b)^x$, where a represents the initial value of the function and b represents the rate of decay, expressed as a decimal. Because $f(0) = 86$, the value of a must be 86. Because the value of $f(x)$ decreases by 80% for each 1-unit increase in x , the value of b must be $(1 - 0.80)$, or 0.2. Therefore, the function f can be defined by $f(x) = 86(0.2)^x$. Substituting 2 for x in this function yields $f(2) = 86(0.2)^2$, which is equivalent to $f(2) = 86(0.04)$, or $f(2) = 3.44$. Either 3.44 or 86/25 may be entered as the correct answer.

Alternate approach: It's given that $f(0) = 86$ and that for each increase in x by 1, the value of $f(x)$ decreases by 80%. Therefore, when $x = 1$, the value of $f(x)$ is $(100 - 80)\%$, or 20%, of 86, which can be expressed as $(0.20)(86)$. Since $(0.20)(86) = 17.2$, the value of $f(1)$ is 17.2. Similarly, when $x = 2$, the value of $f(x)$ is 20% of 17.2, which can be expressed as $(0.20)(17.2)$. Since $(0.20)(17.2) = 3.44$, the value of $f(2)$ is 3.44. Either 3.44 or 86/25 may be entered as the correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$M = 1,800(1.02)^t$

The equation above models the number of members, M, of a gym t years after the gym opens. Of the following, which equation models the number of members of the gym q quarter years after the gym opens?

Answer

- A. $M = 1,800(1.02)^{\frac{q}{4}}$
- B. $M = 1,800(1.02)^{4q}$
- C. $M = 1,800(1.005)^{4q}$
- D. $M = 1,800(1.082)^q$

Correct Answer: A

Rationale

Choice A is correct. In 1 year, there are 4 quarter years, so the number of quarter years, q, is 4 times the number of years, t ; that is, $q = 4t$. This is equivalent to $t = \frac{q}{4}$, and substituting this into the expression for M in terms of t gives $M = 1,800(1.02)^{\frac{q}{4}}$.

Choices B and D are incorrect and may be the result of incorrectly using $t = 4q$ instead of $q = 4t$. (Choices B and D are nearly the same since 1.02^{4q} is equivalent to $(1.02^4)^q$, which is approximately 1.082^q .) Choice C is incorrect and may be the result of incorrectly using $t = 4q$ and unnecessarily dividing 0.02 by 4.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$D = T - \frac{9}{25}(100 - H)$$

The formula above can be used to approximate the dew point D, in degrees Fahrenheit, given the temperature T, in degrees Fahrenheit, and the relative humidity of H percent, where $H > 50$. Which of the following expresses the relative humidity in terms of the temperature and the dew point?

Answer

- A. $H = \frac{25}{9}(D - T) + 100$
- B. $H = \frac{25}{9}(D - T) - 100$
- C. $H = \frac{25}{9}(D + T) + 100$
- D. $H = \frac{25}{9}(D + T) - 100$

Correct Answer: A

Rationale

Choice A is correct. It's given that $D = T - \frac{9}{25}(100 - H)$. Solving this formula for H expresses the relative humidity in terms of the temperature and the dew point. Subtracting T from both sides of this equation yields $D - T = -\frac{9}{25}(100 - H)$. Multiplying both sides by $-\frac{25}{9}$ yields $-\frac{25}{9}(D - T) = 100 - H$. Subtracting 100 from both sides yields $-\frac{25}{9}(D - T) - 100 = -H$. Multiplying both sides by -1 results in the formula $\frac{25}{9}(D - T) + 100 = H$.

Choices B, C, and D are incorrect and may result from errors made when rewriting the given formula.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

Which of the following is equivalent to $\left(a + \frac{b}{2}\right)^2$?

Answer

A. $a^2 + \frac{b^2}{2}$

B. $a^2 + \frac{b^2}{4}$

C. $a^2 + \frac{ab}{2} + \frac{b^2}{2}$

D. $a^2 + ab + \frac{b^2}{4}$

Correct Answer: D

Rationale

Choice D is correct. The expression $\left(a + \frac{b}{2}\right)^2$ can be rewritten as $\left(a + \frac{b}{2}\right)\left(a + \frac{b}{2}\right)$. Using the distributive property, the expression yields $\left(a + \frac{b}{2}\right)\left(a + \frac{b}{2}\right) = a^2 + \frac{ab}{2} + \frac{ab}{2} + \frac{b^2}{4}$. Combining like terms gives $a^2 + ab + \frac{b^2}{4}$.

Choices A, B, and C are incorrect and may result from errors using the distributive property on the given expression or combining like terms.

Question ID: 635f54ee

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The surface area of a cube is $6\left(\frac{a}{4}\right)^2$, where a is a positive constant. Which of the following gives the perimeter of one face of the cube?

Answer

- A. $\frac{a}{4}$
- B. a
- C. $4a$
- D. $6a$

Correct Answer: B

Rationale

Choice B is correct. A cube has 6 faces of equal area, so if the total surface area of a cube is $6\left(\frac{a}{4}\right)^2$, then the area of one face is $\left(\frac{a}{4}\right)^2$.

Likewise, the area of one face of a cube is the square of one of its edges; therefore, if the area of one face is $\left(\frac{a}{4}\right)^2$, then the length of one edge of the cube is $\frac{a}{4}$. Since the perimeter of one face of a cube is four times the length of one edge, the perimeter is $4\left(\frac{a}{4}\right) = a$.

Choice A is incorrect because if the perimeter of one face of the cube is $\frac{a}{4}$, then the total surface area of the cube is $6\left(\frac{\frac{a}{4}}{4}\right)^2 = 6\left(\frac{a}{16}\right)^2$,

which is not $6\left(\frac{a}{4}\right)^2$. Choice C is incorrect because if the perimeter of one face of the cube is $4a$, then the total surface area of the cube is

$6\left(\frac{4a}{4}\right)^2 = 6a^2$, which is not $6\left(\frac{a}{4}\right)^2$. Choice D is incorrect because if the perimeter of one face of the cube is $6a$, then the total surface

area of the cube is $6\left(\frac{6a}{4}\right)^2 = 6\left(\frac{3a}{2}\right)^2$, which is not $6\left(\frac{a}{4}\right)^2$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

According to a model, the estimated volume of a certain tree in a forest increases by **13.9%** for every increase of **100** centimeters in the tree's height. The equation $y = ab^{\frac{x}{100}}$ represents this model, where y represents the estimated volume, in cubic centimeters, of this tree in the forest, x represents the tree's height, in centimeters, and a and b are constants. In this equation, what is the value of b ?

Answer

- A. **0.139**
- B. **1**
- C. **1.139**
- D. **13.9**

Correct Answer: C

Rationale

Choice C is correct. It's given that equation $y = ab^{\frac{x}{100}}$ represents the model, where y is the estimated volume, in cubic centimeters, of a certain tree in a forest and x is the tree's height, in centimeters. For an exponential equation of this form to represent a model where the value of y increases as the value of x increases, it must follow that the values of a and b are positive. It's also given that the tree's estimated volume increases by **13.9%** for every increase of **100** centimeters in the tree's height. Therefore, for every **100**-centimeter increase in the tree's height, the estimated volume of the tree is **1.139** times larger. In other words, $ab^{\frac{x+100}{100}} = 1.139ab^{\frac{x}{100}}$, which is equivalent to $ab^{\frac{x}{100}}b^{\frac{100}{100}} = 1.139ab^{\frac{x}{100}}$. Since a and b are positive, dividing both sides of this equation by $ab^{\frac{x}{100}}$ yields $b^{\frac{100}{100}} = 1.139$, or $b = 1.139$. Therefore, the value of b is **1.139**.

Choice A is incorrect. This is the given percent increase in decimal form, not the value of b .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the given percent increase, not the value of b .

Question ID: 1697ffcf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

In the xy -plane, the graph of $y = 3x^2 - 14x$ intersects the graph of $y = x$ at the points $(0, 0)$ and (a, a) . What is the value of a ?

Rationale

The correct answer is 5. The intersection points of the graphs of $y = 3x^2 - 14x$ and $y = x$ can be found by solving the system consisting of these two equations. To solve the system, substitute x for y in the first equation. This gives $x = 3x^2 - 14x$. Subtracting x from both sides of the equation gives $0 = 3x^2 - 15x$. Factoring $3x$ out of each term on the left-hand side of the equation gives $0 = 3x(x - 5)$. Therefore, the possible values for x are 0 and 5. Since $y = x$, the two intersection points are $(0, 0)$ and $(5, 5)$. Therefore, $a = 5$.

Question ID: 42f8e4b4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

One of the factors of $2x^3 + 42x^2 + 208x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

Correct Answer: 8

Rationale

The correct answer is 8. Since each term of the given expression, $2x^3 + 42x^2 + 208x$, has a factor of $2x$, the expression can be rewritten as $2x(x^2) + 2x(21x) + 2x(104)$, or $2x(x^2 + 21x + 104)$. Since the values 8 and 13 have a sum of 21 and a product of 104, the expression $x^2 + 21x + 104$ can be factored as $(x + 8)(x + 13)$. Therefore, the given expression can be factored as $2x(x + 8)(x + 13)$. It follows that the factors of the given expression are 2, x , $x + 8$, and $x + 13$. Of these factors, only $x + 8$ and $x + 13$ are of the form $x + b$, where b is a positive constant. Therefore, the possible values of b are 8 and 13. Thus, the smallest possible value of b is 8.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy -plane passes through the points $(0, 10)$ and $(-2, \frac{325}{36})$. What is the value of ab ?

Answer

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 54
- D. 60

Correct Answer: C

Rationale

Choice C is correct. It’s given that the function h is defined by $h(x) = a^x + b$ and that the graph of $y = h(x)$ in the xy -plane passes through the points $(0, 10)$ and $(-2, \frac{325}{36})$. Substituting 0 for x and 10 for $h(x)$ in the equation $h(x) = a^x + b$ yields $10 = a^0 + b$, or $10 = 1 + b$. Subtracting 1 from both sides of this equation yields $9 = b$. Substituting -2 for x and $\frac{325}{36}$ for $h(x)$ in the equation $h(x) = a^x + 9$ yields $\frac{325}{36} = a^{-2} + 9$. Subtracting 9 from both sides of this equation yields $\frac{1}{36} = a^{-2}$, which can be rewritten as $a^2 = 36$. Taking the square root of both sides of this equation yields $a = 6$ and $a = -6$, but because it’s given that a is a positive constant, a must equal 6. Because the value of a is 6 and the value of b is 9, the value of ab is $(6)(9)$, or 54.

Choice A is incorrect and may result from finding the value of $a^{-2}b$ rather than the value of ab .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from correctly finding the value of a as 6, but multiplying it by the y -value in the first ordered pair rather than by the value of b .

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

x	y
21	−8
23	8
25	−8

The table shows three values of x and their corresponding values of y , where $y = f(x) + 4$ and f is a quadratic function. What is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?

Correct Answer: -2112

Rationale

The correct answer is $-2,112$. It's given that f is a quadratic function. It follows that f can be defined by an equation of the form $f(x) = a(x - h)^2 + k$, where a , h , and k are constants. It's also given that the table shows three values of x and their corresponding values of y , where $y = f(x) + 4$. Substituting $a(x - h)^2 + k$ for $f(x)$ in this equation yields $y = a(x - h)^2 + k + 4$. This equation represents a quadratic relationship between x and y , where $k + 4$ is either the maximum or the minimum value of y , which occurs when $x = h$. For quadratic relationships between x and y , the maximum or minimum value of y occurs at the value of x halfway between any two values of x that have the same corresponding value of y . The table shows that x -values of 21 and 25 correspond to the same y -value, -8 . Since 23 is halfway between 21 and 25, the maximum or minimum value of y occurs at an x -value of 23. The table shows that when $x = 23$, $y = 8$. It follows that $h = 23$ and $k + 4 = 8$. Subtracting 4 from both sides of the equation $k + 4 = 8$ yields $k = 4$. Substituting 23 for h and 4 for k in the equation $y = a(x - h)^2 + k + 4$ yields $y = a(x - 23)^2 + 4 + 4$, or $y = a(x - 23)^2 + 8$. The value of a can be found by substituting any x -value and its corresponding y -value for x and y , respectively, in this equation. Substituting 25 for x and -8 for y in this equation yields $-8 = a(25 - 23)^2 + 8$, or $-8 = a(2)^2 + 8$. Subtracting 8 from both sides of this equation yields $-16 = a(2)^2$, or $-16 = 4a$. Dividing both sides of this equation by 4 yields $-4 = a$. Substituting -4 for a , 23 for h , and 4 for k in the equation $f(x) = a(x - h)^2 + k$ yields $f(x) = -4(x - 23)^2 + 4$. The y -intercept of the graph of $y = f(x)$ in the xy -plane is the point on the graph where $x = 0$. Substituting 0 for x in the equation $f(x) = -4(x - 23)^2 + 4$ yields $f(0) = -4(0 - 23)^2 + 4$, or $f(0) = -4(-23)^2 + 4$. This is equivalent to $f(0) = -2,112$, so the y -intercept of the graph of $y = f(x)$ in the xy -plane is $(0, -2,112)$. Thus, the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane is $-2,112$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. The model indicates the object has an initial height of **10** feet above the ground and reaches its maximum height of **1,034** feet above the ground **8** seconds after being launched. Based on the model, what is the height, in feet, of the object above the ground **10** seconds after being launched?

Answer

- A. **234**
- B. **778**
- C. **970**
- D. **1,014**

Correct Answer: C

Rationale

Choice C is correct. It's given that a quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. This quadratic function can be defined by an equation of the form $f(x) = a(x - h)^2 + k$, where $f(x)$ is the height of the object x seconds after it was launched, and a , h , and k are constants such that the function reaches its maximum value, k , when $x = h$. Since the model indicates the object reaches its maximum height of **1,034** feet above the ground **8** seconds after being launched, $f(x)$ reaches its maximum value, **1,034**, when $x = 8$. Therefore, $k = 1,034$ and $h = 8$. Substituting **8** for h and **1,034** for k in the function $f(x) = a(x - h)^2 + k$ yields $f(x) = a(x - 8)^2 + 1,034$. Since the model indicates the object has an initial height of **10** feet above the ground, the value of $f(x)$ is **10** when $x = 0$. Substituting **0** for x and **10** for $f(x)$ in the equation $f(x) = a(x - 8)^2 + 1,034$ yields $10 = a(0 - 8)^2 + 1,034$, or $10 = 64a + 1,034$. Subtracting **1,034** from both sides of this equation yields $64a = -1,024$. Dividing both sides of this equation by **64** yields $a = -16$. Therefore, the model can be represented by the equation $f(x) = -16(x - 8)^2 + 1,034$. Substituting **10** for x in this equation yields $f(10) = -16(10 - 8)^2 + 1,034$, or $f(10) = 970$. Therefore, based on the model, **10** seconds after being launched, the height of the object above the ground is **970** feet.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The equation $\frac{x^2+6x-7}{x+7} = ax+d$ is true for all $x \neq -7$, where a and d are integers. What is the value of $a+d$?

Answer

- A. -6
- B. -1
- C. 0
- D. 1

Correct Answer: C

Rationale

Choice C is correct. Since the expression x^2+6x-7 can be factored as $(x+7)(x-1)$, the given equation can be rewritten as $\frac{(x+7)(x-1)}{x+7} = ax+d$. Since $x \neq -7$, $x+7$ is also not equal to 0, so both the numerator and denominator of $\frac{(x+7)(x-1)}{x+7}$ can be divided by $x+7$. This gives $x-1 = ax+d$. Equating the coefficient of x on each side of the equation gives $a = 1$. Equating the constant terms gives $d = -1$. The sum is $1+(-1) = 0$.

Choice A is incorrect and may result from incorrectly simplifying the equation. Choices B and D are incorrect. They are the values of d and a, respectively, not $a+d$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

Which of the following expressions is equivalent to $\frac{x^2-2x-5}{x-3}$?

Answer

- A. $x - 5 - \frac{20}{x-3}$
- B. $x - 5 - \frac{10}{x-3}$
- C. $x + 1 - \frac{8}{x-3}$
- D. $x + 1 - \frac{2}{x-3}$

Correct Answer: D

Rationale

Choice D is correct. The numerator of the given expression can be rewritten in terms of the denominator, $x - 3$, as follows:

$x^2 - 2x - 5 = x^2 - 3x + x - 3 - 2$, which is equivalent to $x(x - 3) + (x - 3) - 2$. So the given expression is equivalent to $\frac{x(x - 3) + (x - 3) - 2}{x - 3} = \frac{x(x - 3)}{x - 3} + \frac{x - 3}{x - 3} - \frac{2}{x - 3}$. Since the given expression is defined for $x \neq 3$, the expression can be rewritten as $x + 1 - \frac{2}{x - 3}$.

Long division can also be used as an alternate approach. Choices A, B, and C are incorrect and may result from errors made when dividing the two polynomials or making use of structure.

Question ID: 9a1f1941

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$|4x - 3| = -9$

How many solutions does the given equation have?

Answer

- A. Zero
- B. One
- C. Two
- D. Infinitely many

Correct Answer: A

Rationale

Choice A is correct. It's given that $|4x - 3| = -9$. By the definition of absolute value, the left-hand side of this equation, $|4x - 3|$, must be nonnegative. Since the right-hand side of this equation, -9 , is negative, it follows that no value of x can make this equation true; therefore, the given equation has zero solutions.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 911c415b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$(7532 + 100y^2) + 10(10y^2 - 110)$

The expression above can be written in the form $ay^2 + b$, where a and b are constants. What is the value of $a + b$?

Rationale

The correct answer is 6632. Applying the distributive property to the expression yields $(7532 + 100y^2) + (100y^2 - 1100)$. Then adding together $7532 + 100y^2$ and $100y^2 - 1100$ and collecting like terms results in $200y^2 + 6432$. This is written in the form $ay^2 + b$, where $a = 200$ and $b = 6432$. Therefore $a + b = 200 + 6432 = 6632$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

Which expression represents the product of $(x^{-6}y^3z^5)$ and $(x^4z^5 + y^8z^{-7})$?

Answer

- A. $x^{-2}z^{10} + y^{11}z^{-2}$
- B. $x^{-2}z^{10} + x^{-6}z^{-2}$
- C. $x^{-2}y^3z^{10} + y^8z^{-7}$
- D. $x^{-2}y^3z^{10} + x^{-6}y^{11}z^{-2}$

Correct Answer: D

Rationale

Choice D is correct. The product of $(x^{-6}y^3z^5)$ and $(x^4z^5 + y^8z^{-7})$ can be represented by the expression $(x^{-6}y^3z^5)(x^4z^5 + y^8z^{-7})$. Applying the distributive property to this expression yields $(x^{-6}y^3z^5)(x^4z^5) + (x^{-6}y^3z^5)(y^8z^{-7})$, or $x^{-6}x^4y^3z^5z^5 + x^{-6}y^3y^8z^5z^{-7}$. This expression is equivalent to $x^{-6+4}y^3z^{5+5} + x^{-6}y^{3+8}z^{5-7}$, or $x^{-2}y^3z^{10} + x^{-6}y^{11}z^{-2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: f89e1d6f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

If $a = c + d$, which of the following is equivalent to the expression $x^2 - c^2 - 2cd - d^2$?

Answer

- A. $(x + a)^2$
- B. $(x - a)^2$
- C. $(x + a)(x - a)$
- D. $x^2 - ax - a^2$

Correct Answer: C

Rationale

Choice C is correct. Factoring -1 from the second, third, and fourth terms gives $x^2 - c^2 - 2cd - d^2 = x^2 - (c^2 + 2cd + d^2)$. The expression $c^2 + 2cd + d^2$ is the expanded form of a perfect square: $c^2 + 2cd + d^2 = (c + d)^2$. Therefore, $x^2 - (c^2 + 2cd + d^2) = x^2 - (c + d)^2$. Since $a = c + d$, $x^2 - (c + d)^2 = x^2 - a^2$. Finally, because $x^2 - a^2$ is the difference of squares, it can be expanded as $x^2 - a^2 = (x + a)(x - a)$.

Choices A and B are incorrect and may be the result of making an error in factoring the difference of squares $x^2 - a^2$. Choice D is incorrect and may be the result of incorrectly combining terms.

Question ID: 2c5c22d0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$y = x^2 + 3x - 7$
 $y - 5x + 8 = 0$

How many solutions are there to the system of equations above?

Answer

- A. There are exactly 4 solutions.
- B. There are exactly 2 solutions.
- C. There is exactly 1 solution.
- D. There are no solutions.

Correct Answer: C

Rationale

Choice C is correct. The second equation of the system can be rewritten as $y = 5x - 8$. Substituting $5x - 8$ for y in the first equation gives $5x - 8 = x^2 + 3x - 7$. This equation can be solved as shown below:

$x^2 + 3x - 7 - 5x + 8 = 0$

$x^2 - 2x + 1 = 0$

$(x - 1)^2 = 0$

$x = 1$

Substituting 1 for x in the equation $y = 5x - 8$ gives $y = -3$. Therefore, $(1, -3)$ is the only solution to the system of equations.

Choice A is incorrect. In the xy-plane, a parabola and a line can intersect at no more than two points. Since the graph of the first equation is a parabola and the graph of the second equation is a line, the system cannot have more than 2 solutions. Choice B is incorrect. There is a single ordered pair (x,y) that satisfies both equations of the system. Choice D is incorrect because the ordered pair $(1, -3)$ satisfies both equations of the system.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$\frac{4x^2}{x^2-9} - \frac{2x}{x+3} = \frac{1}{x-3}$$

What value of x satisfies the equation above?

Answer

- A. -3
- B. $-\frac{1}{2}$
- C. $\frac{1}{2}$
- D. 3

Correct Answer: C

Rationale

Choice C is correct. Each fraction in the given equation can be expressed with the common denominator x^2-9 . Multiplying $\frac{2x}{x+3}$ by $\frac{x-3}{x-3}$ yields $\frac{2x^2-6x}{x^2-9}$, and multiplying $\frac{1}{x-3}$ by $\frac{x+3}{x+3}$ yields $\frac{x+3}{x^2-9}$. Therefore, the given equation can be written as $\frac{4x^2}{x^2-9} - \frac{2x^2-6x}{x^2-9} = \frac{x+3}{x^2-9}$. Multiplying each fraction by the denominator results in the equation $4x^2 - (2x^2 - 6x) = x + 3$, or $2x^2 + 6x = x + 3$. This equation can be solved by setting a quadratic expression equal to 0, then solving for x. Subtracting $x + 3$ from both sides of this equation yields $2x^2 + 5x - 3 = 0$. The expression $2x^2 + 5x - 3$ can be factored, resulting in the equation $(2x - 1)(x + 3) = 0$. By the zero product property, $2x - 1 = 0$ or $x + 3 = 0$. To solve for x in $2x - 1 = 0$, 1 can be added to both sides of the equation, resulting in $2x = 1$. Dividing both sides of this equation by 2 results in $x = \frac{1}{2}$. Solving for x in $x + 3 = 0$ yields $x = -3$. However, this value of x would result in the second fraction of the original equation having a denominator of 0. Therefore, $x = -3$ is an extraneous solution. Thus, the only value of x that satisfies the given equation is $x = \frac{1}{2}$.

Choice A is incorrect and may result from solving $x + 3 = 0$ but not realizing that this solution is extraneous because it would result in a denominator of 0 in the second fraction. Choice B is incorrect and may result from a sign error when solving $2x - 1 = 0$ for x. Choice D is incorrect and may result from a calculation error.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = 4x^2 - 50x + 126$

The given equation defines the function f . For what value of x does $f(x)$ reach its minimum?

Correct Answer: 25/4, 6.25

Rationale

The correct answer is $\frac{25}{4}$. The given equation can be rewritten in the form $f(x) = a(x - h)^2 + k$, where a , h , and k are constants. When $a > 0$, h is the value of x for which $f(x)$ reaches its minimum. The given equation can be rewritten as $f(x) = 4(x^2 - \frac{50}{4}x) + 126$, which is equivalent to $f(x) = 4(x^2 - \frac{50}{4}x + (\frac{50}{8})^2 - (\frac{50}{8})^2) + 126$. This equation can be rewritten as $f(x) = 4((x - \frac{50}{8})^2 - (\frac{50}{8})^2) + 126$, or $f(x) = 4(x - \frac{50}{8})^2 - 4(\frac{50}{8})^2 + 126$, which is equivalent to $f(x) = 4(x - \frac{25}{4})^2 - \frac{121}{4}$. Therefore, $h = \frac{25}{4}$, so the value of x for which $f(x)$ reaches its minimum is $\frac{25}{4}$. Note that 25/4 and 6.25 are examples of ways to enter a correct answer.

Question ID: 9f2ecade

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$h(x) = x^3 + ax^2 + bx + c$

The function h is defined above, where a, b, and c are integer constants. If the zeros of the function are -5 , 6, and 7, what is the value of c ?

Rationale

The correct answer is 210. Since -5 , 6, and 7 are zeros of the function, the function can be rewritten as $h(x) = (x + 5)(x - 6)(x - 7)$. Expanding the function yields $h(x) = x^3 - 8x^2 - 23x + 210$. Thus, $a = -8$, $b = -23$, and $c = 210$. Therefore, the value of c is 210.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$64x^2 + bx + 25 = 0$

In the given equation, b is a constant. For which of the following values of b will the equation have more than one real solution?

Answer

- A. -91
- B. -80
- C. 5
- D. 40

Correct Answer: A

Rationale

Choice A is correct. A quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has either no real solutions, exactly one real solution, or exactly two real solutions. That is, for the given equation to have more than one real solution, it must have exactly two real solutions. When the value of the discriminant, or $b^2 - 4ac$, is greater than 0, the given equation has exactly two real solutions. In the given equation, $64x^2 + bx + 25 = 0$, $a = 64$ and $c = 25$. Therefore, the given equation has exactly two real solutions when $(b)^2 - 4(64)(25) > 0$, or $b^2 - 6,400 > 0$. Adding 6,400 to both sides of this inequality yields $b^2 > 6,400$. Taking the square root of both sides of $b^2 > 6,400$ yields two possible inequalities: $b < -80$ or $b > 80$. Of the choices, only choice A satisfies $b < -80$ or $b > 80$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

If a and c are positive numbers, which of the following is equivalent to $\sqrt{(a+c)^3} \cdot \sqrt{a+c}$?

Answer

- A. $a + c$
- B. $a^2 + c^2$
- C. $a^2 + 2ac + c^2$
- D. a^2c^2

Correct Answer: C

Rationale

Choice C is correct. Using the property that $\sqrt{x}\sqrt{y} = \sqrt{xy}$ for positive numbers x and y, with $x = (a + c)^3$ and $y = a + c$, it follows that $\sqrt{(a+c)^3} \cdot \sqrt{a+c} = \sqrt{(a+c)^4}$. By rewriting $(a + c)^4$ as $((a + c)^2)^2$, it is possible to simplify the square root expression as follows:

$$\sqrt{((a+c)^2)^2} = (a+c)^2 = a^2 + 2ac + c^2.$$

Choice A is incorrect and may be the result of $\sqrt{(a+c)^3} \div \sqrt{(a+c)}$. Choice B is incorrect and may be the result of incorrectly rewriting $(a + c)^2$ as $a^2 + c^2$. Choice D is incorrect and may be the result of incorrectly applying properties of exponents.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$\frac{1}{7}(x - k)(16 - x^2) = 0$

In the given equation, k is a positive constant. The sum of the solutions to the equation is **59**. What is the value of k ?

Correct Answer: 59

Rationale

The correct answer is **59**. It's given that in the equation $\frac{1}{7}(x - k)(16 - x^2) = 0$, k is a positive constant. Multiplying both sides of this equation by **7** yields $(x - k)(16 - x^2) = 0$. By the zero product property, it follows that $x - k = 0$ or $16 - x^2 = 0$. Adding k to both sides of the equation $x - k = 0$ yields $x = k$. Adding x^2 to both sides of the equation $16 - x^2 = 0$ yields $16 = x^2$. Taking the square root of both sides of this equation yields $4 = x$ and $-4 = x$. Therefore, the solutions to the given equation are k , **4**, and **-4**. It's given that the sum of the solutions to the equation is **59**. Thus, $k + 4 + (-4) = 59$, or $k = 59$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$-14(5x - 2)^2 + 6(5x - 3)^2$$

The given expression can be rewritten as $\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6}$, where a , b , and c are constants. What is the value of $a + b + c$?

Answer

- A. **−612**
- B. −306
- C. −102
- D. −17

Correct Answer: A

Rationale

Choice A is correct. Expanding each squared binomial in the given expression yields $-14(25x^2 - 20x + 4) + 6(25x^2 - 30x + 9)$. Applying the distributive property to both terms of this expression yields $(-350x^2 + 280x - 56) + (150x^2 - 180x + 54)$. Combining like terms in this expression yields $-200x^2 + 100x - 2$. Since the given expression can be rewritten as $\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6}$, it follows that $\frac{a}{6} = -200$, $\frac{b}{6} = 100$, and $\frac{c}{6} = -2$. Multiplying both sides of each of these equations by 6 yields $a = -1,200$, $b = 600$, and $c = -12$, respectively. Substituting $-1,200$ for a , 600 for b , and -12 for c in the expression $a + b + c$ yields $-1,200 + 600 - 12$, or -612 .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $\frac{a}{6} + \frac{b}{6} + \frac{c}{6}$, not $a + b + c$.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$p(t) = 90,000(1.06)^t$$

The given function p models the population of Lowell t years after a census. Which of the following functions best models the population of Lowell m months after the census?

Answer

- A. $r(m) = \frac{90,000}{12}(1.06)^m$
- B. $r(m) = 90,000\left(\frac{1.06}{12}\right)^m$
- C. $r(m) = 90,000\left(\frac{1.06}{12}\right)^{\frac{m}{12}}$
- D. $r(m) = 90,000(1.06)^{\frac{m}{12}}$

Correct Answer: D

Rationale

Choice D is correct. It’s given that the function p models the population of Lowell t years after a census. Since there are **12** months in a year, m months after the census is equivalent to $\frac{m}{12}$ years after the census. Substituting $\frac{m}{12}$ for t in the equation $p(t) = 90,000(1.06)^t$ yields $p(\frac{m}{12}) = 90,000(1.06)^{\frac{m}{12}}$. Therefore, the function r that best models the population of Lowell m months after the census is $r(m) = 90,000(1.06)^{\frac{m}{12}}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The population of a town is currently 50,000, and the population is estimated to increase each year by 3% from the previous year. Which of the following equations can be used to estimate the number of years, t, it will take for the population of the town to reach 60,000 ?

Answer

- A. $50,000 = 60,000(0.03)^t$
- B. $50,000 = 60,000(3)^t$
- C. $60,000 = 50,000(0.03)^t$
- D. $60,000 = 50,000(1.03)^t$

Correct Answer: D

Rationale

Choice D is correct. Stating that the population will increase each year by 3% from the previous year is equivalent to saying that the population each year will be 103% of the population the year before. Since the initial population is 50,000, the population after t years is given by $50,000(1.03)^t$. It follows that the equation $60,000 = 50,000(1.03)^t$ can be used to estimate the number of years it will take for the population to reach 60,000.

Choice A is incorrect. This equation models how long it will take the population to decrease from 60,000 to 50,000, which is impossible given the growth factor. Choice B is incorrect and may result from misinterpreting a 3% growth as growth by a factor of 3. Additionally, this equation attempts to model how long it will take the population to decrease from 60,000 to 50,000. Choice C is incorrect and may result from misunderstanding how to model percent growth by multiplying the initial amount by a factor greater than 1.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$\frac{\sqrt[3]{n^{17}p^5}}{5n^2(\sqrt[8]{p^8})}$$

For all positive values of n and p , the given expression is equivalent to which of the following?

I. $\frac{\left(n^{\frac{14}{3}}\right)\left(p^{\frac{5}{3}}\right)}{5np}$

II. $\frac{\sqrt[3]{n^{11}p^2}}{5}$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Correct Answer: C

Rationale

Choice C is correct. For positive values of a , $\sqrt[N]{a} = a^{\frac{1}{N}}$. Therefore, the given expression, $\frac{\sqrt[3]{n^{17}p^5}}{5n^2(\sqrt[8]{p^8})}$, can be rewritten as $\frac{(n^{17}p^5)^{\frac{1}{3}}}{5n^2(p^8)^{\frac{1}{8}}}$. Applying properties of exponents, this can be rewritten as $\frac{n^{\frac{17}{3}}p^{\frac{5}{3}}}{5n^2p^{\frac{5}{8}}}$, or $\frac{n^{\frac{17}{3}}p^{\frac{5}{3}}}{5n^2p}$. Dividing the numerator and the denominator of this expression by n yields $\frac{n^{\frac{14}{3}}p^{\frac{5}{3}}}{5np}$, which is statement I. Dividing the numerator and the denominator of this expression by np yields $\frac{n^{\frac{11}{3}}p^{\frac{2}{3}}}{5}$. This expression is equivalent to $\frac{(n^{11}p^2)^{\frac{1}{3}}}{5}$. It follows that this expression is equivalent to $\frac{\sqrt[3]{n^{11}p^2}}{5}$, which is statement II. Therefore, for all positive values of n and p , the given expression is equivalent to I and II.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$h(x) = -16x^2 + 100x + 10$$

The quadratic function above models the height above the ground h , in feet, of a projectile x seconds after it had been launched vertically. If $y = h(x)$ is graphed in the xy -plane, which of the following represents the real-life meaning of the positive x -intercept of the graph?

Answer

- A. The initial height of the projectile
- B. The maximum height of the projectile
- C. The time at which the projectile reaches its maximum height
- D. The time at which the projectile hits the ground

Correct Answer: D

Rationale

Choice D is correct. The positive x -intercept of the graph of $y = h(x)$ is a point (x,y) for which $y = 0$. Since $y = h(x)$ models the height above the ground, in feet, of the projectile, a y -value of 0 must correspond to the height of the projectile when it is 0 feet above ground or, in other words, when the projectile is on the ground. Since x represents the time since the projectile was launched, it follows that the positive x -intercept, $(x,0)$, represents the time at which the projectile hits the ground.

Choice A is incorrect and may result from misidentifying the y -intercept as a positive x -intercept. Choice B is incorrect and may result from misidentifying the y -value of the vertex of the graph of the function as an x -intercept. Choice C is incorrect and may result from misidentifying the x -value of the vertex of the graph of the function as an x -intercept.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

x	$f(x)$
1	a
2	a^5
3	a^9

For the exponential function f , the table above shows several values of x and their corresponding values of $f(x)$, where a is a constant greater than 1. If k is a constant and $f(k) = a^{29}$, what is the value of k ?

Rationale

The correct answer is 8. The values of $f(x)$ for the exponential function f shown in the table increase by a factor of a^4 for each increase of 1 in x . This relationship can be represented by the equation $f(x) = a^{4x+b}$, where b is a constant. It's given that when $x = 2$, $f(x) = a^5$. Substituting 2 for x and a^5 for $f(x)$ into $f(x) = a^{4x+b}$ yields $a^5 = a^{4(2)+b}$. Since $4(2) + b = 5$, it follows that $b = -3$. Thus, an equation that defines the function f is $f(x) = a^{4x-3}$. It follows that the value of k such that $f(k) = a^{29}$ can be found by solving the equation $4k - 3 = 29$, which yields $k = 8$.

Question ID: f25a34aa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x + 22$ centimeters, and the height of the triangle is $x - 10$ centimeters. What is the value of x ?

Correct Answer: 110

Rationale

The correct answer is 110. The area of a triangle is equal to one half of the product of the length of the base of the triangle and the height of the triangle. It's given that the length of the base of the triangle is $2x + 22$ centimeters and the height of the triangle is $x - 10$ centimeters; therefore, its area is $\frac{1}{2}(2x + 22)(x - 10)$ square centimeters. It's also given that the area of the triangle is equal to x^2 square centimeters. Therefore, it follows that $\frac{1}{2}(2x + 22)(x - 10) = x^2$. This equation can be rewritten as $(x + 11)(x - 10) = x^2$, or $x^2 + x - 110 = x^2$. Subtracting x^2 from both sides of this equation yields $x - 110 = 0$. Adding 110 to both sides of this equation yields $x = 110$. Therefore, the value of x is 110.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The functions g and h are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

- I. $g(x) = 18(1.16)(1.4)^{x+2}$
- II. $h(x) = 18(1.4)^{x+4}$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Correct Answer: D

Rationale

Choice D is correct. A function defined by an equation in the form $f(x) = a(b)^{x+h}$, where a , b , and h are positive constants and $x \geq 0$, has a minimum value of $f(0)$. It's given that function g is defined by $g(x) = 18(1.16)(1.4)^{x+2}$, which is equivalent to $g(x) = 20.88(1.4)^{x+2}$. Substituting 0 for x in this equation yields $g(0) = 20.88(1.4)^{0+2}$, or $g(0) = 40.9248$. Therefore, the minimum value of $g(x)$ is 40.9248, so $g(x) = 18(1.16)(1.4)^{x+2}$ doesn't display its minimum value as a constant or coefficient. It's also given that function h is defined by $h(x) = 18(1.4)^{x+4}$. Substituting 0 for x in this equation yields $h(0) = 18(1.4)^{0+4}$, or $h(0) = 69.1488$. Therefore, the minimum value of $h(x)$ is 69.1488, so $h(x) = 18(1.4)^{x+4}$ doesn't display its minimum value as a constant or coefficient. Therefore, neither I nor II displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

What is the minimum value of the function f defined by $f(x) = (x - 2)^2 - 4$?

Answer

- A. -4
- B. -2
- C. 2
- D. 4

Correct Answer: A

Rationale

Choice A is correct. The given quadratic function f is in vertex form, $f(x) = (x - h)^2 + k$, where (h, k) is the vertex of the graph of $y = f(x)$ in the xy -plane. Therefore, the vertex of the graph of $y = f(x)$ is $(2, -4)$. In addition, the y -coordinate of the vertex represents either the minimum or maximum value of a quadratic function, depending on whether the graph of the function opens upward or downward. Since the leading coefficient of f (the coefficient of the term $(x - 2)^2$) is 1 , which is positive, the graph of $y = f(x)$ opens upward. It follows that at $x = 2$, the minimum value of the function f is -4 .

Choice B is incorrect and may result from making a sign error and from using the x -coordinate of the vertex. Choice C is incorrect and may result from using the x -coordinate of the vertex. Choice D is incorrect and may result from making a sign error.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

Let the function p be defined as $p(x) = \frac{(x - c)^2 + 160}{2c}$, where c is a constant. If $p(c) = 10$, what is the value of $p(12)$?

Answer

- A. 10.00
- B. 10.25
- C. 10.75
- D. 11.00

Correct Answer: D

Rationale

Choice D is correct. The value of p(12) depends on the value of the constant c, so the value of c must first be determined. It is given that p(c) = 10. Based on the definition of p, it follows that:

$$p(c) = \frac{(c - c)^2 + 160}{2c} = 10$$

$$\frac{160}{2c} = 10$$

$$2c = 16$$

$$c = 8$$

This means that $p(x) = \frac{(x - 8)^2 + 160}{16}$ for all values of x. Therefore:

$$p(12) = \frac{(12 - 8)^2 + 160}{16}$$

$$= \frac{16 + 160}{16}$$

$$= 11$$

Choice A is incorrect. It is the value of p(8), not p(12). Choices B and C are incorrect. If one of these values were correct, then x = 12 and the selected value of p(12) could be substituted into the equation to solve for c. However, the values of c that result from choices B and C each result

in $p(c) < 10$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$r(t) = 63t - 2t^2$

The function r is defined by the given equation. The function s is defined by $s(t) = r(t) + 1$. Which expression represents the maximum value of $s(t)$?

Answer

- A. $1 + \left(\frac{63}{2}\right)^2$
- B. $1 + 2\left(\frac{63}{4}\right)^2$
- C. $1 - 2\left(\frac{63}{4}\right)^2$
- D. $1 - \left(\frac{63}{2}\right)^2$

Correct Answer: B

Rationale

Choice B is correct. It's given that $s(t) = r(t) + 1$. Since $r(t) = 63t - 2t^2$, it follows that $s(t) = 63t - 2t^2 + 1$, or $s(t) = -2t^2 + 63t + 1$. For a quadratic function s defined by an equation of the form $s(t) = a(t - h)^2 + k$, where a , h , and k are constants and a is negative, the value of $s(t)$ reaches its maximum, k , when the value of t is h . The equation $s(t) = -2t^2 + 63t + 1$ can be rewritten in the form $s(t) = a(t - h)^2 + k$ by completing the square. The equation $s(t) = -2t^2 + 63t + 1$ is equivalent to $s(t) = -2\left(t^2 - \frac{63}{2}t\right) + 1$, or $s(t) = -2\left(t^2 - \frac{63}{2}t + \left(\frac{63}{4}\right)^2 - \left(\frac{63}{4}\right)^2\right) + 1$. This equation can be rewritten as $s(t) = -2\left(\left(t - \frac{63}{4}\right)^2 - \left(\frac{63}{4}\right)^2\right) + 1$, or $s(t) = -2\left(t - \frac{63}{4}\right)^2 + 1 + 2\left(\frac{63}{4}\right)^2$. This equation is in the form $s(t) = a(t - h)^2 + k$, where $a = -2$, $h = \frac{63}{4}$, and $k = 1 + 2\left(\frac{63}{4}\right)^2$. Therefore, the maximum value of $s(t)$ is $1 + 2\left(\frac{63}{4}\right)^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 58b109d4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$x^2 + y + 7 = 7$$

$$20x + 100 - y = 0$$

The solution to the given system of equations is (x, y) . What is the value of x ?

Correct Answer: -10

Rationale

The correct answer is -10 . Adding y to both sides of the second equation in the given system yields $20x + 100 = y$. Substituting $20x + 100$ for y in the first equation in the given system yields $x^2 + 20x + 100 + 7 = 7$. Subtracting 7 from both sides of this equation yields $x^2 + 20x + 100 = 0$. Factoring the left-hand side of this equation yields $(x + 10)(x + 10) = 0$, or $(x + 10)^2 = 0$. Taking the square root of both sides of this equation yields $x + 10 = 0$. Subtracting 10 from both sides of this equation yields $x = -10$. Therefore, the value of x is -10 .